Probability & Statistics
Open-Book Binder v2

Midterm (UTS) edition · Devore 9E, Sessions 1–12 · English with *(istilah Indonesia)* · Casio fx-991ID

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The prof's exam format (from Quiz 1)

Both class versions had the same six question types, in this order. Expect the midterm to look similar. Sessions 9–12 material (pdf, exponential) was not on Quiz 1 but may appear.

#	Question type	Method page	Solved example
1	Name the variable type and measurement scale *(jenis variabel & skala pengukuran)*	**p.8**	**p.31**
2	15 data values: mean, median, five-number summary, IQR, s, CV, stem-and-leaf, boxplot + outliers	**p.9**	**p.33**
3	Contingency table: joint, union, conditional, independence	**p.12**	**p.36**
4	Binomial: "x% fail, n chosen", at most / at least / from a to b, mean and SD	**p.15**	**p.37**
5	Poisson: "rata-rata 4 per jam", rescale for 30 minutes	**p.15**	**p.38**
6	Normal: "rata-rata 50, **varians** 15", P(<), P(between), P(>), the 97% cut-off	**p.20**	**p.39**

> ✔ In the exam: (1) find the question type in Tab I, (2) copy the layout of the solved example in Tab R, (3) check the number with DIST mode, (4) write the formula and substitution on your paper.

Sources: lecture decks (Anom Yudistira, based on Devore 9E), readings in the course repo, class whiteboard photos, your OneNote notes, and Quiz 1 LA40/LB40. Every number was recomputed by computer.

G · Glossary: Indonesian question words → meaning → math

The most dangerous words are the ones that change **≤ vs <** or **which side** of the distribution you need. They are marked ★.

Inequalities

Indonesian	English	Math
★ paling sedikit / setidaknya / minimal / sekurang-kurangnya	at least	$X \geq k$
★ paling banyak / maksimal / sebanyak-banyaknya	at most	$X \leq k$
★ tidak lebih dari	not more than	$X \leq k$
★ tidak kurang dari	not less than	$X \geq k$
lebih dari	more than	$X > k$
kurang dari	less than	$X < k$
tepat / sebanyak / ada (k)	exactly	$X = k$
antara a dan b	between a and b	$a \leq X \leq b$*
ada 2–4 / dari 2 sampai 4	from 2 to 4	$2 \leq X \leq 4$
sedikitnya satu / minimal satu	at least one	$1 - \mathrm{P}(X = 0)$
tidak ada / tidak satu pun	none	$X = 0$
semua	all	$X = n$

*For a **continuous** X, ≤ and < give the same answer. For a **discrete** X (Binomial, Poisson) they do not. If "antara" is ambiguous for a discrete X, write down which you assumed, e.g. "diasumsikan inklusif" (assumed inclusive).

Instructions

Hitunglah / Tentukan / Carilah	Calculate / Determine / Find
Buktikan bahwa ... merupakan pdf	Prove that ... is a valid pdf (check f ≥ 0 and ∫ = 1)
Lengkapi tabel	Complete the table
Diketahui	Given / It is known that
Jika / apabila	If
Jelaskan / Interpretasikan	Explain / Interpret
Gambarkan	Draw / sketch
secara acak	at random
Berapa peluang ...	What is the probability ...
Berapa banyak cara ...	How many ways ... (counting)

Probability words

peluang / probabilitas	probability
kejadian	event
ruang sampel	sample space S
percobaan	experiment
saling lepas / saling asing	mutually exclusive ($A \cap B = \emptyset$)
saling bebas / independen	independent ($\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$)
peluang bersyarat	conditional probability $\mathrm{P}(A \mid B)$
★ jika diketahui / dengan syarat / diberikan	given that → the condition goes **after** the bar
gabungan (∪) / irisan (∩)	union / intersection
komplemen / bukan	complement A^c
peluang gabungan (joint)	joint probability $\mathrm{P}(A \cap B)$
peluang marginal	marginal probability (row/column total ÷ grand total)
himpunan kosong	empty set ∅
permutasi / kombinasi	permutation (order matters) / combination
★ tanpa pengembalian	without replacement → Hypergeometric
dengan pengembalian	with replacement → Binomial
dadu / koin / kartu / bola / kotak / guci	die / coin / card / ball / box / urn
mata dadu / selisih / jumlah	die face / difference / sum
cacat / rusak / baik	defective / broken / good
lulus / tidak lulus (TL) / mengulang	pass / fail / retake

Statistics & distribution words

skala pengukuran: nominal / ordinal / interval / rasio	measurement scale (see **p.8**)
variabel kategorik / numerik	categorical (qualitative) / numerical (quantitative)
koefisien variasi (CV)	coefficient of variation = $s/\bar{x} \times 100\%$
ringkasan lima angka	five-number summary
diagram batang-daun / diagram kotak	stem-and-leaf / boxplot
kestabilan / konsistensi	stability → compare CV (smaller = more stable)
setidaknya	at least (≥), same as paling sedikit
variabel random / acak (diskrit / kontinyu)	random variable (discrete / continuous)
fungsi peluang (pmf / pdf)	probability function (pmf / pdf)
fungsi kumulatif (cdf)	cumulative distribution function F(x)
rata-rata / nilai harapan	mean / expected value $\mu = E(X)$
varians / ragam	variance σ^2
simpangan baku / standar deviasi	standard deviation σ
median / modus / kuartil / persentil	median / mode / quartile / percentile
jangkauan	range
pencilan	outlier
populasi / sampel	population (N, μ, σ) / sample ($n, \bar{x}, s$)
★ laju / rata-rata kejadian per (jam, hari, halaman)	rate λ → Poisson (count) or Exponential (time)
waktu tunggu / umur / masa pakai	waiting time / lifetime → Exponential
distribusi normal baku	standard normal $Z \sim N(0,1)$
seragam	uniform
sukses / gagal	success / failure (Binomial)
ketebalan / berat bersih / tinggi badan / suhu	thickness / net weight / height / temperature

⚠ **TRAP** Indonesian decimals use a **comma** and thousands use a **dot**: "0,4" = 0.4 and "15.015" = 15015 (the class normal-distribution example). Read every number twice.

I · Problem-type index

Find the phrase in the question, then go to the page.

The question says …	Method	Page
Data and variables		
"jenis variabel", "skala pengukuran", list of variables	qualitative/quantitative + nominal/ordinal/interval/ratio	p.8
"mean, median, modus" of a data list	$\bar{x} = \sum x/n$; median at position $0.5(n+1)$	p.8
"varians / standar deviasi" of data	s^2 with n - 1; STAT mode **sx**	p.9
"coefficient of variation / koefisien variasi", "lebih stabil"	CV = s / x̄ × 100%	p.9
"five number summary", "IQR", "kuartil"	Q1, Q3 at positions 0.25(n+1), 0.75(n+1)	p.9
"stem and leaf", "boxplot", "outlier / pencilan"	fences Q1 - 1.5·IQR and Q3 + 1.5·IQR	p.10
"at least what proportion within k SD", "mound-shaped"	Chebyshev $1 - 1/k^2$ / empirical 68–95–99.7	p.9
Probability		
a two-way table of counts	joint = cell/N; marginal = total/N; conditional = cell/given total	p.12
"… **atau** …" (or)	$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$	p.11
"**jika diketahui** …" (given that)	$\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$	p.13
"apakah … independen / saling bebas?"	compare $\mathrm{P}(A \mid B)$ with $\mathrm{P}(A)$	p.13
machines with % share and % defect, "it is defective; which machine?"	total probability, then Bayes	p.13
"berapa banyak cara", committee, arrangement	mn rule, P_r^n, C_r^n	p.11
Distributions		
table of x and P(x), unknown k, E(X), Var(X)	$\sum p = 1$; $\sum xp$; $\sum x^2 p - \mu^2$	p.14
"**x% gagal/cacat**, dipilih n secara acak"	**Binomial** B(n, p)	p.15
"**rata-rata λ per** jam/menit/halaman"	**Poisson**(λ), rescale λ to the interval	p.15
"tanpa pengembalian" from a small lot	Hypergeometric	p.16
"rata-rata μ dan **varians** σ²" (bell-shaped measurement)	**Normal**: σ = √varians, then z	p.20
"agar 97% … dalam waktu tersebut", "top 5%"	work backwards: $x = \mu + z\sigma$	p.21
Binomial with large n, "pendekatan normal"	continuity correction ±0.5	p.21
"buktikan f(x) pdf", "tentukan k", P(a ≤ X ≤ b) from f(x)	$f \geq 0$, $\int f = 1$; integrate	p.17
cdf F(x), median/percentile of a pdf, E and Var of a pdf	$F = \int f$; solve $F(\eta) = p$; $\int xf$	p.17
"seragam" on [A, B]	Uniform	p.19
"waktu tunggu / umur", "rata-rata waktu …"	**Exponential**, λ = 1/mean	p.22
Y = g(X)	cdf method	p.22

D · Distribution picker

X is a COUNT (0, 1, 2, …)

- n trials, success/failure, same p, independent → **Binomial** · p.15
- n drawn *without replacement* from a small lot → **Hypergeometric** · p.16
- events per time/space at an average rate, no fixed n → **Poisson** · p.15
- a custom experiment (dice, coins, a given table) → build the pmf · p.14

X is a MEASUREMENT (time, weight, temperature)

- formula f(x) given → integrate · p.17
- equally likely on [A, B] → **Uniform** · p.19
- bell-shaped, mean and variance given → **Normal** · p.20
- waiting time / lifetime → **Exponential**, λ = 1/mean · p.22

Formula table

Distribution	pmf / pdf	Values	Mean	Variance	Calculator
Binomial B(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	0 … n	np	$np(1-p)$	Binomial PD / CD
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!}$	0, 1, 2, …	λ	λ	Poisson PD / CD
Hypergeometric	$\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$	0 … min(n, M)	$n\frac{M}{N}$	$\frac{N-n}{N-1}\,n\frac{M}{N}\left(1-\frac{M}{N}\right)$	nCr
Uniform[A, B]	$\frac{1}{B-A}$	A ≤ x ≤ B	$\frac{A+B}{2}$	$\frac{(B-A)^2}{12}$	by hand
Normal N(μ, σ²)	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	all real x	μ	σ^2	Normal CD / Inverse
Exponential(λ)	$\lambda e^{-\lambda x}$	x ≥ 0	$1/\lambda$	$1/\lambda^2$	e^x key

Geometric and Negative Binomial: see p.16.

CDFs

Uniform $F(x)=\frac{x-A}{B-A}$ · Exponential $F(x)=1-e^{-\lambda x}$

Normal $F(x)=\Phi\left(\frac{x-\mu}{\sigma}\right)$ · Binomial/Poisson: CD key or Tab T

Rules for any X

$\mathrm{E}(aX+b)=a\,\mathrm{E}(X)+b$ · $\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}(X)$

$\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$ · SD = √Var

C · Calculator (fx-991ID)

Need	Keys	Inputs	Check value
Binomial P(X = x)	MODE DIST → Binomial PD → Var	x, N, p	3, 8, 0.4 → 0.27869
Binomial P(X ≤ x)	DIST → Binomial CD → Var	x, N, p	3, 8, 0.4 → 0.59409
Poisson P(X = x) / P(X ≤ x)	DIST → Poisson PD / CD → Var	x, μ (= λ)	3, 4 → 0.19537 / 0.43347
Normal P(a ≤ X ≤ b)	DIST → Normal CD	Lower, Upper, **σ**, μ	485, 515, 8, 500 → 0.93921
Normal x for a left area	DIST → Inverse Normal	Area, **σ**, μ	0.97, √15, 50 → 57.284
Data: $\bar{x}$, s	MODE STAT → 1-VAR, enter data, AC, SHIFT 1 → Var	use **sx**, not σx	2, 9, 11, 5, 6 → sx 3.5071
Integral	∫▭dx in COMP, X = ALPHA)	f(x), lower, upper	$\int_3^4(0.075x+0.2)$ → 0.4625
Solve F(x) = p	SHIFT CALC	equation, start guess	median of a pdf
nCr / nPr / x! / e^x	SHIFT ÷ / SHIFT × / SHIFT x^{-1} / SHIFT ln		20C2 = 190

⚠ **TRAP Normal CD and Inverse Normal take σ, not the variance.** "Varians 15" → enter σ = √15 = 3.873.

⚠ **TRAP One-sided normal:** P(X ≤ b): Lower = -1×10^{99}. P(X ≥ a): Upper = 1×10^{99}. Inverse Normal is left-tail only: "top 5%" → Area 0.95.

▣ **CALCULATOR Converting CD output** (discrete): P(X < k) = CD(k-1) · P(X ≥ k) = 1 - CD(k-1) · P(X > k) = 1 - CD(k) · P(a ≤ X ≤ b) = CD(b) - CD(a-1)

Store unrounded values with STO A, B … and round only the final answer to 4 decimals.

1 · Overview & descriptive statistics (Sessions 1–2, Devore Ch. 1)

Vocabulary

- **Population** *(populasi)*: all measurements of interest. Size N, mean μ, variance σ^2. **Sample** *(sampel)*: a subset. Size n, mean $\bar{x}$, variance s^2.
- **Variable** *(variabel: karakteristik / alat ukur)* vs **data** *(hasil pengukuran)*.
- **Experimental unit**: the object measured (person, light bulb).
- Uni-/bi-/multivariate: 1, 2, or more variables per unit.

Types of variables

- **Qualitative / categorical** *(non angka)*: gender, major, hair colour, ethnic group. A numeric code (1 = M, 2 = F) is still qualitative, and a student ID is qualitative too.
- **Quantitative** *(angka)*:
  - **Discrete** *(bilangan bulat)*: countable values (number of phones, semester, # of units).
  - **Continuous**: any value in an interval (body temperature 36,2°, GPA, time, weight).

Graphs

- Qualitative: **bar chart**, **pie chart**; frequency table: item | freq | relative freq (= f/n) | percent.
- Quantitative: **dotplot**, **stem-and-leaf**, **(relative-frequency) histogram**, time series (line chart).
- Histogram classes: width ≈ (max - min) / #classes, rounded **up** to a convenient number. Example: (70 - 26)/6 = 7.33 → width 8, starting at 25.

Stem-and-leaf *(batang-daun)*: (1) sort small → large *(urutkan)*; (2) split into stem | leaf, e.g. tens | ones; (3) give a key, e.g. "6 | 5 = 65" or for 1 decimal: "18 | 6 = 18.6".
Board example: 17.2, 18.6, 18.8, 18.8, 20.1, 21.3, 23.5 → `17|2 18|688 19| 20|1 21|3 22| 23|5`

Shapes *(bentuk distribusi)*

Mound-shaped, symmetric	mean ≈ median
Skewed right (long right tail, a few unusually **large** values)	**mean > median**
Skewed left (long left tail, a few unusually **small** values)	**mean < median**
Bimodal	two local peaks

Measurement scales *(skala pengukuran)*: Nominal · Ordinal · Interval · Ratio

Quiz 1 question 1 asked for **both** the variable type and the scale of every variable. Answer with two labels for each variable, e.g. "kuantitatif kontinu, rasio".

Scale	What it means	Allowed	Examples (Computer Engineering)
Nominal	Categories with **no order**. Numbers used as labels are still nominal.	count, mode, =, ≠	jenis OS, jenis sensor, jenis mikrokontroler, protokol komunikasi, merek, status ON/OFF, gagal/tidak gagal, NIM, ID perangkat
Ordinal	Categories with a **natural order**, but the gaps between them are not equal or measurable.	+ median, <, >, ranking	kondisi Baik/Sedang/Rusak, suhu CPU Normal/Tinggi, grade A/B/C, tingkat kepuasan 1–5, prioritas rendah/sedang/tinggi, peringkat
Interval	Numbers with equal gaps, but **zero is arbitrary** (0 does not mean "none"). Differences make sense, ratios do not.	+ mean, SD, differences	suhu in **°C or °F**, tahun (calendar year), jam/tanggal (clock time)
Ratio	Numbers with equal gaps and a **true zero** (0 = none). "Twice as much" makes sense.	everything, including CV and ratios	waktu respons, kecepatan, daya (W), tegangan (V), amplitudo (mV), berat, panjang, jumlah perangkat/kegagalan, persentase error, akurasi (%), suhu in **Kelvin**

Decision steps

1. Are the values words or labels? → **qualitative / kategorik**.
   Is there a natural order? Yes → **ordinal**. No → **nominal**.
2. Are the values numbers you can add and average? → **quantitative / numerik**.
   Counted (0, 1, 2 …) → **discrete**. Measured → **continuous**.
3. Does 0 mean "none of it"? Yes → **ratio**. No (0°C still has heat) → **interval**.

⚠ **TRAP "Tingkat …" (level of …)** is ambiguous. "Tingkat akurasi 97,5%" is a number, so it is ratio. "Tingkat akurasi tinggi/sedang/rendah" is ordinal. When unsure, write the assumption, e.g. "diasumsikan dalam persen (%) → rasio".

⚠ **TRAP Counts are ratio, not interval.** "Jumlah kegagalan" = 0 means no failures, a true zero. Temperature in °C is the classic interval example.

Answer template	Variable	Type *(jenis)*	Scale *(skala)*	Reason (one line)
Write one row per variable	Waktu respons sensor (ms)	Kuantitatif kontinu	Rasio	measured; 0 ms = no time
	Jenis protokol komunikasi	Kualitatif	Nominal	labels (I2C, SPI, UART) with no order

The full worked answer for all 15 Quiz 1 variables is on **p.31**.

Measures of centre *(ukuran pemusatan)*

Mean: $\bar{x} = \frac{\sum x_i}{n}$ (sample); $\mu = \frac{\sum x_i}{N}$ (population)

Median: the middle value after sorting. Its **position** is $0.5(n+1)$. If the position ends in .5, average the two neighbours.

Mode *(modus)*: the most frequent value. A data set can have 2 modes (bimodal) or none.

Example (slides). 2, 4, 9, 8, 6, 5, 3 → sorted 2 3 4 5 6 8 9, position 0.5(8) = 4th → median = 5.
2, 4, 9, 8, 6, 5 → sorted 2 4 5 6 8 9, position 3.5 → median = (5+6)/2 = 5.5.

Robustness (board). 2, 3, 4, 5, 6: x̄ = 4, median 4. Add an extreme value 1000: x̄ = 170, median = 4.5. The median is **robust** *(stabil)*, so use it for skewed data.

Measures of variability *(ukuran penyebaran)*

Range *(jangkauan)* R = max - min.

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N} \quad \text{(population)}$$

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1} \quad \text{(sample)}$$

$s = \sqrt{s^2}$ is always taken positive (≥ 0). It is in the original units, while s^2 is in squared units.

We divide by n - 1 for a sample because x̄ is estimated from the same data. This makes s^2 a better (unbiased) estimate of σ^2.

Example (slides): petals 5, 12, 6, 8, 14.
n = 5, Σx = 45, x̄ = 9, Σx^2 = 25+144+36+64+196 = 465

$$s^2 = \frac{465 - 45^2/5}{4} = \frac{465 - 405}{4} = 15, \; s = 3.873.$$

Range = 14 - 5 = 9.

▣ **CALCULATOR** STAT 1-VAR: enter 5, 12, 6, 8, 14 → sx = 3.8730 → sx^2 = 15. Do **not** use σx (= 3.4641) for a sample.

Range approximation: R ≈ 4s to 6s, so s ≈ R/4 (a rough check).

Coefficient of variation *(koefisien variasi, CV)*

$$CV = \frac{s}{\bar{x}} \times 100\% \quad \text{(sample)} \qquad CV = \frac{\sigma}{\mu} \times 100\% \quad \text{(population)}$$

CV is the SD as a percentage of the mean. It has **no unit**, so you can compare spread between data sets with different units or very different means. **Smaller CV = more stable / consistent** *(lebih stabil, lebih konsisten)*.
Use it only for ratio-scale data with a positive mean.

Example. Sensor A: x̄ = 34 mV, s = 18.36 mV → CV = 18.36/34 × 100% = 54%. Sensor B: x̄ = 30 mV, s = 3 mV → CV = 10%. Sensor B is far more stable.

✔ Use the **sample** SD s (calculator sx), unless the question says the data are a whole population. Show both numbers in the formula: "CV = 18,3555 / 33,9333 × 100% = 54,09%".

Rough guide often used in engineering reports: CV below 10% is low, 10–30% moderate, above 30% high variability. State it as an interpretation, not a rule.

Chebyshev (Tchebysheff) vs the Empirical rule

Chebyshev, any shape: at least $1 - \frac{1}{k^2}$ of the values lie within k SDs of the mean (k ≥ 1).
k = 2 → at least 3/4 = 75%; k = 3 → at least 8/9 ≈ 88.9%.
To find k: $k = \frac{\text{distance from mean}}{s}$.

Empirical rule, mound-shaped only: about 68% within μ ± σ, 95% within μ ± 2σ, 99.7% within μ ± 3σ.
Tails: P(above μ + 2σ) ≈ 2.5%; P(above μ + σ) ≈ 16%.

Example (slides). Task time: mean 12.8 min, SD 1.7, mound-shaped. Proportion longer than 16.2 min? 16.2 = 12.8 + 2(1.7) is 2 SDs above, so ≈ (1 - 0.95)/2 = **0.025**.

Relative standing

z-score: $z = \frac{x - \bar{x}}{s}$. |z| ≤ 2 is not unusual; |z| > 3 suggests an outlier. **p-th percentile**: p% of the values lie below it.

Quartiles, IQR, five-number summary

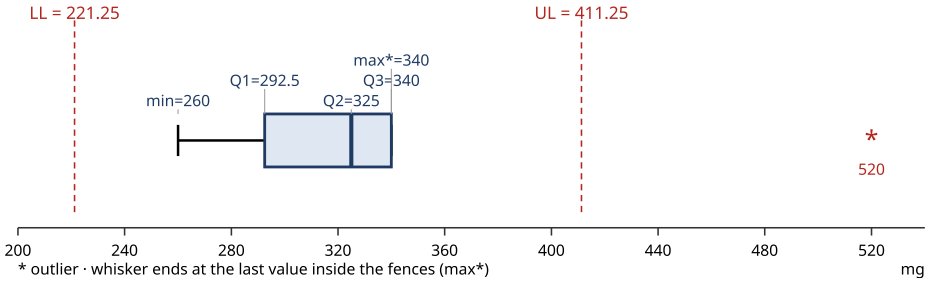
Sort the data first *(urutkan)*. Positions (slides' rule): Q_1 at $0.25(n+1)$ · median Q_2 at $0.5(n+1)$ · Q_3 at $0.75(n+1)$
If a position is fractional, **interpolate**: position 4.75 → $x_{(4)} + 0.75\,(x_{(5)} - x_{(4)})$.
n = 15 (both Quiz 1 versions): Q1 = 4th value, median = 8th, Q3 = 12th. No interpolation needed.
$IQR = Q_3 - Q_1$
Five-number summary *(ringkasan lima angka)*: Min, Q1, Median, Q3, Max.

Fences *(batas pencilan)*:
LL = $Q_1 - 1.5\,IQR$ · UL = $Q_3 + 1.5\,IQR$
A value below LL or above UL is an **outlier** *(pencilan)*, drawn as *.
Whiskers go to the **most extreme values that are not outliers**, not to the fences.
Shape: median closer to Q1 and a long right side → **skewed right** (mean > median). Median closer to Q3 → skewed left.

Drawing the boxplot (diagram kotak)

1. Draw a number line with a scale that covers the min, max and both fences.
2. Draw a box from Q1 to Q3 and a vertical line at the median.
3. Compute LL and UL. Mark them with dashed lines (optional, but it shows your check).
4. Draw each whisker to the last data value inside the fences.
5. Mark each outlier with *.
6. Write the conclusion: "Terdapat outlier: 65, 67, 75 (> UL = 36,5)" or "Tidak terdapat outlier karena semua data berada di antara LL dan UL".

Slides: sodium in 8 cheeses 260 290 300 320 330 340 340 520
Q1 position 2.25 → 292.5 · median 4.5 → 325 · Q3 position 6.75 → 340 · IQR 47.5 · LL 221.25 · UL 411.25 → **520 is an outlier**, the upper whisker stops at 340.



Slides: shoe prices, n = 18 40 60 65 65 65 68 68 70 70 70 70 70 70 74 75 75 90 95. Q1 position 4.75 → 65; Q3 position 14.25 → 74 + 0.25(75 - 74) = 74.25; median 70; IQR 9.25; fences 51.125 and 88.125 → **40, 90 and 95 are outliers**. (The slide gets 75.25 because its list leaves out the 40.)

⚠ **TRAP** Different books and calculators use different quartile rules. Use the **0.25(n+1)** rule and write the position you used.

2 · Probability (Sessions 3–4, Devore Ch. 2)

Sample spaces and events

- **Experiment** *(percobaan)*: a process that produces an outcome. **Sample space S** *(ruang sampel)*: the set of all simple outcomes.
- **Event** *(kejadian)*: a subset of S, written with a capital letter.
- $A \cup B$: A or B or both *(gabungan)*. $A \cap B$: both *(irisan)*. A^c (also A′): not A *(komplemen)*.
- **Mutually exclusive** *(saling lepas)*: $A \cap B = \emptyset$, so P(A ∩ B) = 0.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.

Axioms and basic rules

1. $\mathrm{P}(A) \geq 0$ · 2. $\mathrm{P}(S) = 1$ · 3. If $A_1, A_2, \ldots$ are mutually exclusive: $\mathrm{P}(\cup A_i) = \sum \mathrm{P}(A_i)$
→ $0 \leq \mathrm{P}(A) \leq 1$, $\mathrm{P}(\emptyset) = 0$, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$
Equally likely outcomes: $\mathrm{P}(A) = \dfrac{n(A)}{n(S)}$

Relative-frequency interpretation: P(A) is the long-run proportion n(A)/n as n → ∞.

Board: n(S) = 100 people, n(A) = 25 wear glasses → P(A) = 25/100 = 0.25.

Addition rule, complement, "at least one"

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

Mutually exclusive: $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$
Three events: $\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(AB) - \mathrm{P}(AC) - \mathrm{P}(BC) + \mathrm{P}(ABC)$

"At least one" ***(sedikitnya satu)***: $\mathrm{P}(\geq 1) = 1 - \mathrm{P}(\text{none})$
If the trials are independent with success prob p each: $1 - (1-p)^n$.
Example: 3 independent alarms each detect with 0.85: $1 - 0.15^3 = 0.9966$.

Example (slides): 120 students. A = Javanese (50), B = female (60), both = 30. $\mathrm{P}(A \cup B) = \frac{50+60-30}{120} = \frac{80}{120} = \frac{2}{3}$.

Counting techniques

Situation	Formula	Example
mn rule (product rule): k stages with $n_1, n_2, \ldots$ ways	$n_1 n_2 \cdots n_k$	Toss 2 dice: 6 × 6 = 36. Three coins: $2^3 = 8$.
Permutation: choose r of n, **order matters** (arrange, rank, code, president/secretary)	$P_r^n = \dfrac{n!}{(n-r)!}$	3-digit lock code from {1,2,3,4}, no repeats: 4P3 = 24. Arrange 5 parts: 5! = 120.
Combination: choose r of n, **order does not matter** (committee, hand, sample)	$C_r^n = \binom{n}{r} = \dfrac{n!}{r!(n-r)!}$	3 of 5 people for a subcommittee: 5C3 = 10.
With repetition allowed, order matters	n^r	4-digit PIN: 10^4
Arrange n objects with identical groups $n_1, n_2, \ldots$	$\dfrac{n!}{n_1!\, n_2! \cdots}$	"STATISTIK": 9!/(2!·3!·2!) for S, T, I

Example (slides): box of 6 M&Ms, 4 red and 2 green, choose 2. P(exactly one red) = $\dfrac{\binom{4}{1}\binom{2}{1}}{\binom{6}{2}} = \dfrac{8}{15} = 0.5333$. Order does not matter, so use combinations in both the numerator and the denominator.

✔ To decide between a permutation and a combination, ask: "If I swap two chosen items, is it a different result?" If yes, it is a permutation. If no, it is a combination.

Contingency tables: joint, marginal, conditional *(tabel kontingensi)*

This was the most-practised type on the class boards. Always write the table **with totals** first.

Age \ Health insurance	Yes	No	Total
18–34	750	170	920
35 & older	950	130	1080
Total	1700	300	2000

Joint (cell ÷ grand total): $\mathrm{P}(18\text{–}34 \cap \mathrm{Yes}) = 750/2000 = 0.375$
Marginal (row or column total ÷ grand total): $\mathrm{P}(\mathrm{No}) = 300/2000 = 0.15$, $\mathrm{P}(18\text{–}34) = 920/2000 = 0.46$
Conditional (cell ÷ total of the *given* row or column):
$\mathrm{P}(\mathrm{No} \mid 18\text{–}34) = 170/920 = 0.1848$
$\mathrm{P}(\mathrm{No} \mid 35{+}) = 130/1080 = 0.1204$
$\mathrm{P}(18\text{–}34 \mid \mathrm{No}) = 170/300 = 0.5667$

Reading P(⋯ | ⋯): the left side is what is asked *(ditanya / dicari)*. The right side is what is known, the condition *(diketahui / syarat)*.
The denominator is always the total of the **condition**.

Sports example (class, Anderson p.191, millions)
Walking: M 28.7, F 57.7, total 86.4. All: M 123.3, F 161.8, total 285.1.
P(walking) = 86.4/285.1 = 0.3031
P(walking | female) = 57.7/161.8 = 0.3566
P(woman | walker) = 57.7/86.4 = 0.6678. This is **conditional**, not joint.
P(woman ∩ walker) = 57.7/285.1 = 0.2024

✔ Independence test from a table: compare $\mathrm{P}(\mathrm{No} \mid 18\text{–}34) = 0.1848$ with $\mathrm{P}(\mathrm{No}) = 0.15$. They differ, so age and insurance are **not independent**.

Conditional probability *(peluang bersyarat)*

$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}, \qquad \mathrm{P}(B) > 0$$

$\mathrm{P}(A \mid B) \neq \mathrm{P}(B \mid A)$ in general, but $\mathrm{P}(A \cap B) = \mathrm{P}(B \cap A)$.
$\mathrm{P}(A^c \mid B) = 1 - \mathrm{P}(A \mid B)$

Example (slides): 5 M&Ms, 2 red and 3 blue, draw 2 without replacement.
P(2nd red | 1st blue) = 2/4 = 1/2; P(2nd red | 1st red) = 1/4. The answer depends on the 1st draw, so the events are **dependent**.

Independence *(saling bebas)* vs mutually exclusive *(saling lepas)*

A and B are **independent** if any one of these holds (then all do):
1. $\mathrm{P}(A \mid B) = \mathrm{P}(A)$ · 2. $\mathrm{P}(B \mid A) = \mathrm{P}(B)$ · 3. $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$
If A and B are independent, so are A^c and B, A and B^c, and A^c and B^c.

⚠ **TRAP** Mutually exclusive events with P(A), P(B) > 0 are **never** independent: P(A ∩ B) = 0 ≠ P(A)P(B). "Saling lepas" is about overlap. "Saling bebas" is about one event giving no information about the other.

Multiplication rule

$\mathrm{P}(A \cap B) = \mathrm{P}(A)\ \mathrm{P}(B \mid A) = \mathrm{P}(B)\ \mathrm{P}(A \mid B)$
Independent: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$
Chain: $\mathrm{P}(A \cap B \cap C) = \mathrm{P}(A)\,\mathrm{P}(B \mid A)\,\mathrm{P}(C \mid A \cap B)$

Example (slides): 10% high-risk. 3 people:
P(exactly one high-risk) = HNN + NHN + NNH = $3(0.1)(0.9)^2$ = **0.243**. This is the Binomial B(3, 0.1) at x = 1.
49% female, P(H | F) = 0.08 → P(H ∩ F) = 0.49 × 0.08 = **0.0392**.

Law of total probability

If $B_1, \ldots, B_k$ are mutually exclusive and exhaustive (a partition: exactly one happens):

$$\mathrm{P}(A) = \sum_{i=1}^{k} \mathrm{P}(B_i)\ \mathrm{P}(A \mid B_i)$$

Draw a **tree**: the first branches are B_i with $\mathrm{P}(B_i)$, the second branches are A | B_i. Multiply along each path, then add the paths that end in A.

Bayes' rule

$$\mathrm{P}(B_j \mid A) = \frac{\mathrm{P}(B_j)\ \mathrm{P}(A \mid B_j)}{\sum_i \mathrm{P}(B_i)\ \mathrm{P}(A \mid B_i)} = \frac{\text{the one path}}{\text{all paths to } A}$$

Prior $\mathrm{P}(B_j)$ → posterior $\mathrm{P}(B_j \mid A)$ after observing A.

Example (slides): heart attack risk. P(F) = 0.49, P(M) = 0.51, P(H | F) = 0.08, P(H | M) = 0.12. A randomly selected person is high-risk. P(male)?
P(H) = 0.49(0.08) + 0.51(0.12) = 0.0392 + 0.0612 = 0.1004
$\mathrm{P}(M \mid H) = \frac{0.0612}{0.1004} = \mathbf{0.6096}$; $\mathrm{P}(F \mid H) = 0.3904$. Check: the two add to 1.

Template: machines. Machines A, B, C make 50%, 30%, 20% of the output, with defect rates 2%, 3%, 5%.
P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = 0.029
P(C | D) = 0.010/0.029 = 0.3448 · P(B | D) = 0.3103 · P(A | D) = 0.3448

✔ Bayes in a table: build a table of joint probabilities, $\mathrm{P}(B_i)\,\mathrm{P}(A \mid B_i)$ for A and for A^c. Then every posterior is "cell ÷ column total", exactly like the contingency tables above.

3 · Discrete random variables (Sessions 5–8, Devore Ch. 3)

Random variable *(variabel random)*

A **function** that maps each outcome of an experiment to a number *(fungsi yang memetakan outcome ke bilangan)*. It is written with a capital letter (X), and its values with a lower-case letter (x).

- Pass/fail status: X = {lulus, tidak lulus} → coded {1, 0}. This is a **Bernoulli** random variable.
- Number of phones owned: X = {0, 1, 2, ...} → **discrete**.
- Midterm score 0 ≤ Z ≤ 100 (any real number) → **continuous** (Ch. 4).

pmf and cdf

pmf *(fungsi peluang / probability mass function)*: $p(x) = \mathrm{P}(X = x)$
Valid if: $0 \le p(x) \le 1$ and $\sum_x p(x) = 1$.
cdf *(fungsi kumulatif)*: $F(x) = \mathrm{P}(X \le x) = \sum_{y \le x} p(y)$. It is a step function, non-decreasing, rising from 0 to 1.
$\mathrm{P}(a \le X \le b) = F(b) - F(a^-)$, where a^- is the possible value just below a.
$p(x) = F(x) - F(x^-)$, the size of the jump at x.

Coin tossed 3 times, X = # heads (class)

8 equally likely outcomes, each 1/8: HHH(3) HHT(2) HTH(2) THH(2) HTT(1) THT(1) TTH(1) TTT(0)

x	0	1	2	3
p(x)	1/8	3/8	3/8	1/8
F(x)	1/8	4/8	7/8	1

$$F(x) = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \le x < 1 \\ 4/8 & 1 \le x < 2 \\ 7/8 & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

Table with an unknown constant

p(x): x = 0, 1, 2, 3 → 0.1, k, 2k, 0.3. Use Σ = 1: 0.4 + 3k = 1, so k = 0.2.
Always check that every p(x) is between 0 and 1 afterwards.

Formula pmf: p(x) = c·x for x = 1, 2, 3, 4 → c(1+2+3+4) = 1 → c = 1/10.

Expected value and variance

$$\mu = \mathrm{E}(X) = \sum_x x\,p(x) \qquad \mathrm{E}[h(X)] = \sum_x h(x)\,p(x)$$
$$\sigma^2 = \mathrm{Var}(X) = \sum (x-\mu)^2 p(x) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$$
$$\sigma = \sqrt{\sigma^2}$$
$\mathrm{E}(aX+b) = a\mu + b$, $\mathrm{Var}(aX+b) = a^2\sigma^2$, $\sigma_{aX+b} = |a|\sigma$

Class contoh soal: X = positive difference of 2 dice *(selisih positif 2 mata dadu)*

Count the 36 pairs: X=0: 6, X=1: 10, X=2: 8, X=3: 6, X=4: 4, X=5: 2.

x	0	1	2	3	4	5	Σ
p(x)	6/36	10/36	8/36	6/36	4/36	2/36	1
x·p	0	10/36	16/36	18/36	16/36	10/36	70/36
x^2·p	0	10/36	32/36	54/36	64/36	50/36	210/36

μ = 70/36 = **1.9444**; $E(X^2)$ = 210/36 = 5.8333; σ^2 = 5.8333 - 1.9444^2 = **2.0525**; σ = 1.4327

Fair die (slides): μ = 21/6 = 3.5, $E(X^2)$ = 91/6, σ^2 = 91/6 - 12.25 = 35/12 = 2.9167, σ = 1.7078.

Binomial distribution *(distribusi binomial)*: X ~ B(n, p)

Check all four conditions (class board):
1. each trial has only **2 outcomes** (success/failure)
2. a **fixed number** of trials n
3. trials are **independent**
4. P(success) = p is **known and constant** on every trial

X = number of successes in n trials.

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

$\mu = np$, $\sigma^2 = np(1-p)$ (q = 1 - p)

⚠ **TRAP** Sampling **without replacement** from a small population breaks independence, so use the Hypergeometric. If the sample is ≤ 5% of the population, the Binomial is OK (Devore).

▣ **CALCULATOR** Binomial PD (x, N, p) for P(X = x). Binomial CD for P(X ≤ x). For P(X ≥ k), use 1 - CD(k-1).

Marksman (slides): hits 80%, 5 shots. P(X = 3) = $10(0.8)^3(0.2)^2$ = 0.2048. P(X > 3) = P(4) + P(5) = 0.4096 + 0.3277 = 0.7373. μ = 4, σ = √0.8 = 0.894. Zero hits is (0 - 4)/0.894 = -4.5 SDs from the mean, so it would be very unusual.

Contoh soal: P(pass) = 0,4, 8 students. X ~ B(8, 0.4). class board

1. P(exactly 3 pass) = $\binom{8}{3}(0.4)^3(0.6)^5 = 56(0.064)(0.07776) =$ **0.2787**

2. P(2–4 pass) = P(2) + P(3) + P(4) = 0.2090 + 0.2787 + 0.2322 = **0.7200**

3. P(at most 3) ***(paling banyak 3)*** = P(0) + P(1) + P(2) + P(3) = 0.0168 + 0.0896 + 0.2090 + 0.2787 = **0.5941** (the board's 0.592 comes from rounding too early)

4. P(at least 8) ***(paling sedikit 8)*** = P(X = 8) = $(0.4)^8$ = **0.00066**

Mean = 8(0.4) = 3.2, variance = 8(0.4)(0.6) = 1.92.

x	0	1	2	3	4	5	6	7	8
p(x)	.0168	.0896	.2090	.2787	.2322	.1239	.0413	.0079	.0007

X ~ B(10, 0.25): a) P(X = 5) = $252(0.25)^5(0.75)^5$ = **0.0584** · b) P(X ≥ 4) = 1 - [P(0) + P(1) + P(2) + P(3)] = 1 - (0.0563 + 0.1877 + 0.2816 + 0.2503) = 1 - 0.7759 = **0.2241** · c) P(X = 0) = 0.75^{10} = **0.0563** · d) P(X ≤ 2) = **0.5256** class board

Poisson distribution *(distribusi Poisson)*: X ~ Poisson(λ)

X = **number of events** in a fixed interval of time or space (calls per hour, typos per page, accidents per week, vehicles through a toll gate).
λ = average number of events **in that interval** *(rata-rata kejadian)*.

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

$\mu = \lambda$, $\sigma^2 = \lambda$

Rescale λ: rate α per unit, interval length t → λ = αt.

Approximates the Binomial when n is large and p is small: λ = np (Devore: n > 50, np < 5).

Typos: λ = 3 per page. class board

1. P(X = 5) = $e^{-3}3^5/5!$ = **0.1008**
2. P(X ≤ 3) = $e^{-3}(1 + 3 + 9/2 + 27/6)$ = 0.0498 + 0.1494 + 0.2240 + 0.2240 = **0.6472**
3. P(X ≥ 2) = 1 - P(0) - P(1) = 1 - 0.0498 - 0.1494 = **0.8009**

Accidents (slides): λ = 2 per week. P(X = 1) = $2e^{-2}$ = 0.2707. P(X ≥ 8) = 1 - P(X ≤ 7) = 1 - 0.9989 = 0.0011.
For **2 weeks**, λ = 4.

Hypergeometric distribution: sampling without replacement *(tanpa pengembalian)*

Population of N items, of which M are "successes". Draw n without replacement. X = number of successes in the sample.

$$\mathrm{P}(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

$$\mathrm{E}(X) = n\frac{M}{N},\ \mathrm{Var}(X) = \frac{N-n}{N-1} \cdot n\frac{M}{N}\left(1 - \frac{M}{N}\right)$$

The factor (N - n)/(N - 1) is the finite-population correction. Without it, the variance is the Binomial one with p = M/N.

Batteries (slides). A pack of 8 has 2 defective. Pick 4. Success = a working battery: N = 8, M = 6, n = 4.

P(all 4 work) = $\frac{\binom{6}{4}\binom{2}{0}}{\binom{8}{4}} = \frac{15}{70} = \mathbf{0.2143}$

E(X) = 4(6/8) = 3; Var(X) = (4/7)(4)(0.75)(0.25) = 0.4286.

Geometric and Negative Binomial (listed in the sub-topics; low priority)

Geometric: X = trial number of the **first** success. $\mathrm{P}(X = x) = (1-p)^{x-1}p$, x = 1, 2, ...
$\mathrm{P}(X > k) = (1-p)^k$, $\mathrm{E}(X) = 1/p$, $\mathrm{Var}(X) = (1-p)/p^2$.
Example: p = 0.1. P(first success on the 3rd try) = $0.9^2(0.1)$ = 0.081. P(success within 3 tries) = $1 - 0.9^3$ = 0.271.

Negative Binomial (Devore's form): X = number of **failures** before the r-th success.

$\mathrm{P}(X = x) = \binom{x+r-1}{r-1}p^r(1-p)^x$, $\mathrm{E}(X) = \frac{r(1-p)}{p}$, $\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$.

Example: r = 2, p = 0.2, x = 3 failures: $\binom{4}{1}(0.2)^2(0.8)^3 = 0.0819$.

Other books count total trials X + r. Check which definition the question uses.

Discrete checklist

1. Define X in words *(X = banyaknya ...)* and list its possible values. The class boards always do this.
2. Name the distribution and its parameters: "X ~ B(8, 0.4)", "X ~ Poisson(3)".
3. Translate the words into an inequality (Tab G), and turn it into = / ≤ / complement form.
4. Write the formula, substitute, and compute. Check with DIST.
5. Sanity check: is 0 ≤ P ≤ 1? Is it near the mean, which is where the largest probabilities are?

4 · Continuous random variables (Sessions 9–12, Devore Ch. 4)

X takes any real value in an interval *(bilangan real, dinyatakan dalam interval / range)*: midterm score 0 ≤ x ≤ 100, height 140 ≤ y ≤ 200, room temperature 20 ≤ z ≤ 38.

Probability density function (pdf)

$$\mathrm{P}(a \le X \le b) = \int_a^b f(x)\,dx \quad (\text{area under } f)$$

f is a valid pdf iff *(syarat pdf)*:
(1) $f(x) \ge 0$ for all x, and (2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$.
Consequences: $\mathrm{P}(X = a) = 0$, so $\mathrm{P}(a \le X \le b) = \mathrm{P}(a < X < b) = \mathrm{P}(a \le X < b) = \dots$
f(x) itself is **not** a probability and can be > 1.

"Buktikan ... pdf" / "find c": template

1. State that f(x) ≥ 0 on the support (e.g. "e^(·) > 0 and 0.15 > 0", or check the endpoints of a linear f).
2. Integrate over the support only (f = 0 elsewhere): $\int_{-\infty}^{\infty} f = \int_{\text{support}} f + \int_{\text{elsewhere}} 0$.
3. Show the result = 1 (or solve c from "= 1").
4. Conclude: "∴ f(x) merupakan pdf" (∴ f(x) is a pdf).

⚠ **TRAP** If ∫ f ≠ 1, the function is **not** a pdf. Do not "fix" it by dividing by the total area unless the question says f is proportional to ... (∝). See the class example on **p.19**.

Contoh soal 1: prove that $f(x) = 0.15e^{-0.15(x-0.5)}$ for x ≥ 0.5 (0 otherwise) is a pdf. class board / Devore Ex. 4.5
(time headway between cars)
(1) For x ≥ 0.5, 0.15 > 0 and the exponential is > 0, so f(x) ≥ 0. Elsewhere f = 0 ≥ 0.
(2) $\int_{-\infty}^{\infty} f(x)dx = \int_{0.5}^{\infty} 0.15e^{-0.15(x-0.5)}dx = \left[-e^{-0.15(x-0.5)}\right]_{0.5}^{\infty} = 0 - (-e^0) = 1$ ✓ ∴ f is a pdf.
Extra: P(X ≤ 5) = $\left[-e^{-0.15(x-0.5)}\right]_{0.5}^{5} = 1 - e^{-0.675} = \mathbf{0.4908}$. cdf: $F(x) = 1 - e^{-0.15(x-0.5)}$ for x ≥ 0.5. This is an exponential distribution with λ = 0.15 shifted right by 0.5, so E(X) = 0.5 + 1/0.15 = 7.1667 and Var(X) = 1/0.15² = 44.44.

Cumulative distribution function (cdf) *(fungsi kumulatif)*

$$F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$

$\mathrm{P}(a \le X \le b) = F(b) - F(a) \cdot \mathrm{P}(X > a) = 1 - F(a)$
$f(x) = F'(x)$ wherever the derivative exists
F is continuous and non-decreasing, with F(-∞) = 0 and F(∞) = 1.

Writing a piecewise cdf (support [A, B]): $F(x) = \begin{cases} 0 & x < A \\ \int_A^x f(t)\,dt & A \le x \le B \\ 1 & x > B \end{cases}$
Check: the middle piece must be 0 at x = A and 1 at x = B.

Gravel sales (tons/week): $f(x) = 1.5(1 - x^2)$, 0 ≤ x ≤ 1. slides / Devore Ex. 4.7
$F(x) = \int_0^x 1.5(1 - t^2)dt = 1.5\left(x - \frac{x^3}{3}\right)$ for 0 ≤ x ≤ 1 (0 below, 1 above).
P(X ≤ 0.25) = 1.5(0.25 - 0.005208) = **0.3672**; P(X > 0.75) = 1 - 1.5(0.75 - 0.140625) = **0.0859**; P(0.25 < X < 0.75) = 0.9141 - 0.3672 = **0.5469**
E(X) = $\int_0^1 x \cdot 1.5(1 - x^2)dx = 1.5\left(\frac{1}{2} - \frac{1}{4}\right) = \mathbf{0.375}$; E(X²) = 1.5(1/3 - 1/5) = 0.2; Var = 0.2 - 0.375² = **0.0594**; σ = 0.2437

Percentiles and the median

The (100p)-th percentile η(p) solves $F(\eta(p)) = p$. Median $\tilde{\mu}$: $F(\tilde{\mu}) = 0.5$. Quartiles: F = 0.25 and F = 0.75.
For a symmetric pdf, median = mean = centre of symmetry.

Gravel, 40th percentile: 1.5(η - η³/3) = 0.4 → η³ - 3η + 0.8 = 0 → **η = 0.2735** (SHIFT SOLVE, start at 0.3). Median: 1.5(m - m³/3) = 0.5 → m = 0.3473.

Expected value and variance (continuous)

$$\mu = \mathrm{E}(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

$$\mathrm{E}[h(X)] = \int h(x) f(x)\, dx$$

$$\sigma^2 = \mathrm{Var}(X) = \int (x-\mu)^2 f(x)\, dx = \mathrm{E}(X^2) - \mu^2$$

$\mathrm{E}(X^2) = \int x^2 f(x)\, dx$ · the rules for aX + b are the same as in the discrete case.

✔ Each integral runs over the **support only**. Remember the extra x (or x^2) inside. The most common slip is computing ∫ f(x) dx (= 1) when you meant ∫ x·f(x) dx.

Uniform distribution *(distribusi seragam)*: X ~ U[A, B]

$f(x) = \frac{1}{B-A}$ for A ≤ x ≤ B · $F(x) = \frac{x-A}{B-A}$

$\mathrm{P}(c \le X \le d) = \frac{d-c}{B-A}$ (length ratio)

$\mathrm{E}(X) = \frac{A+B}{2}$, $\mathrm{Var}(X) = \frac{(B-A)^2}{12}$, median = (A + B)/2

Tire imperfection angle (slides): X ~ U[0, 360). P(within 90° of the reference line) = P(0 ≤ X ≤ 90) + P(270 ≤ X < 360) = 90/360 + 90/360 = **0.5**.
Bus every 20 min: wait W ~ U[0, 20]. P(W > 15) = 5/20 = 0.25. E = 10 min, Var = 400/12 = 33.33.

Given f(x) = 0,075x + 0,2 for 3 ≤ x ≤ 5 (0 otherwise). Find a) P(X ≤ 4) b) P(X > 4) c) P(3.5 ≤ X ≤ 4.5) d) P(X > 4.5) e) μ f) σ².

class board, contoh soal 2

Note: the board reads "0,75x", but that integrates to 6.4 over [3, 5], so it is not a valid pdf. With **0,075** the integral is exactly 1, which is clearly what was intended. In class the students divided by 6.4 instead. That gives different numbers, e.g. 0.441 vs 0.4625 for (a).

Validity: f(3) = 0.425 > 0 and f(5) = 0.575 > 0, and f is linear, so f ≥ 0 on [3, 5]. $\int_3^5 (0.075x + 0.2)dx = \left[0.0375x^2 + 0.2x\right]_3^5 = (0.9375 + 1) - (0.3375 + 0.6) = 1$ ✓
Antiderivative G(x) = 0.0375x² + 0.2x. **cdf**: F(x) = 0.0375x² + 0.2x - 0.9375 for 3 ≤ x ≤ 5.

a) P(X ≤ 4)	$\int_3^4 f = G(4) - G(3) = 1.4 - 0.9375$	**0.4625**
b) P(X > 4)	$\int_4^5 f + \int_5^\infty 0 = 1.9375 - 1.4$ (or 1 - 0.4625)	**0.5375**
c) P(3.5 ≤ X ≤ 4.5)	$G(4.5) - G(3.5) = 1.659375 - 1.159375$	**0.5000**
d) P(X > 4.5)	$G(5) - G(4.5) = 1.9375 - 1.659375$	**0.2781**
e) μ	$\int_3^5 x(0.075x + 0.2)dx = \left[0.025x^3 + 0.1x^2\right]_3^5 = (3.125 + 2.5) - (0.675 + 0.9)$	**4.05**
E(X²)	$\int_3^5 x^2(0.075x + 0.2)dx = \left[0.01875x^4 + \frac{0.2}{3}x^3\right]_3^5 = (11.71875 + 8.3333) - (1.51875 + 1.8)$	16.7333
f) σ²	$16.7333 - 4.05^2 = 16.7333 - 16.4025$	**0.3308**
(extra) median	$F(m) = 0.5 \Rightarrow 0.0375m^2 + 0.2m - 1.4375 = 0 \Rightarrow m = \frac{-0.2 + \sqrt{0.04 + 0.215625}}{0.075}$	4.0746

Sanity check: the density increases with x, so the mean (4.05) is a little right of the midpoint 4, and P(X > 4) > 0.5. ✓

Discrete ↔ continuous dictionary

	Discrete	Continuous
probability function	pmf p(x) = P(X = x)	pdf f(x); P(X = x) = 0
total	Σ p(x) = 1	∫ f(x) dx = 1
cdf	$F(x) = \Sigma_{y \le x}\, p(y)$ (steps)	$F(x) = \int_{-\infty}^{x} f$ (smooth)
mean / E(X²)	Σ x p(x) / Σ x² p(x)	∫ x f(x) dx / ∫ x² f(x) dx
≤ vs <	**matters**	does not matter

5 · Normal distribution *(distribusi normal)*

$X \sim N(\mu, \sigma^2)$ with parameters μ (mean, -∞ < μ < ∞) and **σ² (variance) > 0**. Domain (-∞, ∞).

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,dt$. This has no closed form, so we standardize and use the z-table (or Normal CD).

The curve is bell-shaped and symmetric about μ: mean = median = mode. A larger σ makes it wider and flatter.

⚠ **TRAP N(500, 64) means σ² = 64, so σ = 8.** The class writes the variance as the second parameter. Always take the square root before standardizing. If the question says "simpangan baku 8", then σ = 8 directly.

Empirical rule: P(|Z| ≤ 1) = 0.6826, P(|Z| ≤ 2) = 0.9544, P(|Z| ≤ 3) = 0.9974.
Symmetry: Φ(-z) = 1 - Φ(z); P(Z > z) = Φ(-z).

Standard normal Z ~ N(0, 1) *(normal baku)* and how to use the table

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1), \qquad f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Steps (class board):
1. Convert x to z: $z = (x-\mu)/\sigma$
2. Round z to **2 decimals** *(Z hanya boleh 2 desimal)*
3. Read Φ(z) = area to the **left** from Tab T (**p.24**).

Asked	Use
$\mathrm{P}(X < x) = \mathrm{P}(X \le x)$	$\Phi(z)$ directly *(langsung lihat tabel)*
$\mathrm{P}(X > x) = \mathrm{P}(X \ge x)$	$1 - \Phi(z)$
$\mathrm{P}(a < X < b)$	$\Phi(z_b) - \Phi(z_a)$
$\mathrm{P}(X = x)$	**0** (continuous)
z > 3.49 (off the table)	Φ(z) ≈ 1
z < -3.49	Φ(z) ≈ 0

Semiconductor thickness: mean 500 nm, variance 64 → X ~ N(500, 64), σ = 8. Spec is 485–515 nm. class board

1. $\mathrm{P}(485 \le X \le 515)$: $z = \frac{485-500}{8} = -1.875 \to -1.88$, $z = \frac{515-500}{8} = 1.875 \to 1.88$
   = Φ(1.88) - Φ(-1.88) = 0.9699 - 0.0301 = **0.9398** (Normal CD / ∫ gives 0.9392)
2. $\mathrm{P}(X \le 510)$: z = 10/8 = 1.25 → Φ(1.25) = **0.8944**

X ~ N(15015, 3540²), so σ = 3540. class problem "17"

a) P(X > 18000): z = 2985/3540 = 0.84 → 1 - 0.7995 = **0.2005**
b) P(X < 10000): z = -5015/3540 = -1.42 → **0.0778**
c) P(12000 < X < 18000): z = -0.85 and 0.84 → 0.7995 - 0.1977 = **0.6018** (exact 0.6033)
d) P(X ≤ 14000): z = -1015/3540 = -0.29 → **0.3859** (exact 0.3872)

X ~ N(8.46, 0.84): σ² = 0.84, so σ = √0.84 = 0.9165. class board

P(8 ≤ X ≤ 10): z = (10 - 8.46)/0.9165 = 1.68, z = (8 - 8.46)/0.9165 = -0.50 → 0.9535 - 0.3085 = **0.6450**
P(X ≥ 10) = 1 - 0.9535 = **0.0465** · P(X > 15): z = 7.14 > 3.49 → ≈ **0**
(The board used σ = 0.913, which gives z = 1.69, 0.6460 and 0.0455. Use √0.84 = 0.9165.)

Slides: X ~ N(5, 2²): P(X > 7) = 1 - Φ(1) = 0.1587. Ground beef N(1, 0.10²): P(0.80 < X < 0.85) = Φ(-1.5) - Φ(-2) = 0.0668 - 0.0228 = **0.0440**.

Working backwards: percentiles *(mencari nilai x dari peluang)*

1. Turn the statement into a **left** area: "top 1%" or "only 1% exceed" → left area 0.99.
2. Find z in the **body** of the table with area closest to it. If it is halfway between two entries, average the z values.
3. $x = \mu + z\sigma$. In general $\eta(p) = \mu + z_p\,\sigma$.

Left area	z	Left area	z
0.01	-2.33	0.90	1.28
0.025	-1.96	0.95	1.645
0.05	-1.645	0.975	1.96
0.10	-1.28	0.99	2.33
0.25 (Q1)	-0.67	0.995	2.58
0.50	0	0.75 (Q3)	0.67

Slides: the z with 0.05 to its right: left area 0.95 is halfway between 0.9495 (1.64) and 0.9505 (1.65), so z = **1.645**.
Ground beef, a weight that only 1% exceed: z = 2.33 → x = 1 + 2.33(0.10) = **1.233 lb** (Inverse Normal: 1.2326).
Semiconductor: thickness of the thinnest 10%: x = 500 - 1.28(8) = **489.76 nm**.

Normal approximation to the Binomial

If X ~ B(n, p) with **np ≥ 5 and n(1 - p) ≥ 5** (slides: "both greater than 5"; Devore uses 10), then X ≈ N(μ = np, σ^2 = np(1 - p)).
Continuity correction: each integer k covers [k - 0.5, k + 0.5].

Binomial	Normal (with correction)
P(X ≤ k)	P(Y ≤ k + 0.5)
P(X < k)	P(Y ≤ k - 0.5)
P(X ≥ k)	P(Y ≥ k - 0.5)
P(X > k)	P(Y ≥ k + 0.5)
P(X = k)	P(k - 0.5 ≤ Y ≤ k + 0.5)
P(a ≤ X ≤ b)	P(a - 0.5 ≤ Y ≤ b + 0.5)

Slides: n = 30, p = 0.4: np = 12, nq = 18, so the approximation is OK. σ = √7.2 = 2.683. P(X ≤ 10) ≈ P(Y ≤ 10.5) = Φ((10.5 - 12)/2.683) = Φ(-0.56) = **0.2877** (exact Binomial 0.2915).
Batteries: n = 200, p = 0.95: np = 190, nq = 10, so OK. σ = √9.5 = 3.082. P(X ≥ 195) ≈ P(Y ≥ 194.5) = 1 - Φ(1.46) = 1 - 0.9279 = **0.0721** (exact 0.0623; nq is small, so the approximation is rough).

✔ You have Binomial CD in DIST mode. If the question explicitly says "gunakan pendekatan normal" (use the normal approximation), do the approximation. Otherwise give the exact Binomial and mention the approximation as a check.

6 · Exponential distribution *(distribusi eksponensial)*

$X \sim \text{Exp}(\lambda)$, where λ > 0 is the **rate** *(laju)*. Domain x ≥ 0.

$$f(x) = \lambda e^{-\lambda x},\ x \geq 0 \qquad F(x) = 1 - e^{-\lambda x}$$

$$\text{P}(X > x) = e^{-\lambda x} \cdot \text{P}(a \leq X \leq b) = e^{-\lambda a} - e^{-\lambda b}$$

$\text{E}(X) = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$, σ = mean

Median $= \frac{\ln 2}{\lambda} \approx 0.693 \times \text{mean}$ (less than the mean, because it is skewed right)

Used for waiting times, lifetimes, and time between events.

⚠ **TRAP "Rata-rata waktu 90 menit" is the MEAN, so λ = 1/90 per minute.** The class board first wrote "λ = rata-rata", then corrected it to E(X) = 1/λ. If you are given a *rate* ("3 per jam", 3 per hour), then λ = 3 and the mean is 1/3 hour.

Link to Poisson: if events occur at rate λ per unit time (a Poisson process), the time T until the next event is Exp(λ): $\text{P}(T > t) = \text{P}(0 \text{ events in } [0, t]) = e^{-\lambda t}$.
Example: 2 calls per hour. P(wait more than 1 hour) = e^{-2} = 0.1353. P(next call within 15 min) = 1 - e^(-2·0.25) = 0.3935.

The time to finish a statistics exam has mean 90 minutes. X ~ Exp(λ = 1/90). class board

1. "dalam waktu 88 menit" (in 88 minutes). The board wrote P(X = 88) = f(88). For a continuous X, **P(X = 88) = 0**, and f(88) = (1/90)e^(-88/90) = 0.00418 is a *density*, not a probability. The sensible reading is "within 88 minutes": P(X ≤ 88) = 1 - e^(-88/90) = **0.6239**. On the exam, write which reading you used. If they insist on f(88), give 0.00418 and add "(density, P(X = 88) = 0)".
2. P(X < 100) = F(100) = 1 - e^(-100/90) = **0.6708**
3. P(X > 75) = 1 - F(75) = e^(-75/90) = **0.4346**

Median time = 90 ln 2 = 62.38 min.

Memoryless property

$$\text{P}(X \geq t + t_0 \mid X \geq t_0) = \text{P}(X \geq t) = e^{-\lambda t}$$

How long it has already lasted does not matter. Only the **additional** time t counts.

Light bulb (Devore p.172 type): mean life 1000 h. P(it lasts another 500 h given it has already lasted 1000 h) = e^(-500/1000) = **0.6065**, the same as for a new bulb.
Exam: P(X > 120 | X > 60) = e^(-60/90) = 0.5134.

Functions of a continuous random variable: Y = g(X)

CDF method (always works):
1. Find the range of Y.
2. $F_Y(y) = \text{P}(Y \leq y) = \text{P}(g(X) \leq y)$. Rewrite this as an event about X and use F_X.
3. $f_Y(y) = F_Y'(y)$.

Monotone g (shortcut): $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$.

Only E(Y) needed? Use LOTUS: $\text{E}[g(X)] = \int g(x) f_X(x)\, dx$. There is no need for f_Y.

Ex. X ~ U(0, 1), Y = e^X. Range [1, e]. $F_Y(y) = \text{P}(X \leq \ln y) = \ln y$, so $f_Y(y) = 1/y$ on [1, e]. E(Y) = $\int_0^1 e^x\, dx$ = e - 1.
Ex. X ~ U(-1, 1), Y = X^2. $F_Y(y) = \text{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y}$, so $f_Y = \frac{1}{2\sqrt{y}}$ on (0, 1].
Linear: Y = aX + b: E = aμ + b, Var = $a^2\sigma^2$. If X is Normal, Y ~ N(aμ + b, $a^2\sigma^2$).
Ex. (reading): X ~ N(2, 4), Y = 3 - 2X ~ N(-1, 16).

X · Traps

Traps that appeared on Quiz 1

The question says	Trap	Do this
"rata-rata 50 dan **varians 15**"	Using 15 as σ gives P(X > 60) = 0.2525 instead of 0.0049.	σ = √15 = 3.873. Enter σ in Normal CD.
"agar **97%** … dalam waktu tersebut atau lebih cepat"	Looking up 0.03 or using the right tail.	"or faster" = left area 0.97 → z = 1.88 → x = μ + zσ.
"rata-rata 4 per jam … selama **30 menit**"	Keeping λ = 4.	λ = 4 × 0.5 = 2, then P(X = 0) = e^{-2}.
"**setidaknya** 4", "paling sedikit"	1 - CD(4).	P(X ≥ 4) = 1 - CD(**3**).
"berada **dari 2 hingga 5**"	CD(5) - CD(2) drops x = 2.	CD(5) - CD(**1**).
"Windows **atau** rusak"	Adding 60 + 25 counts Windows-rusak twice.	(60 + 25 - 10)/130.
"**jika diketahui** rusak, … Linux?"	Dividing by the grand total.	Divide by the rusak total: 10/25.
"standar deviasi" of 15 data values	σx (÷ n) instead of sx (÷ n - 1).	Use sx; CV uses the same s.
"suhu operasional (°C)"	Calling it ratio.	°C is **interval** (0°C is not "no heat").
"tingkat akurasi / tingkat kesalahan"	Calling every "tingkat" ordinal.	A percentage is ratio. Words like tinggi/rendah are ordinal. State your assumption.
boxplot whiskers	Drawing whiskers to the fences or to the outliers.	Whiskers stop at the last value inside the fences.

Mistakes on the class boards and in your notes

Board / notes say	Correct
f(x) = 0,75x + 0,2 on [3, 5], then divide by 6.4	**0,075x + 0,2** (integrates to 1). Answers on **p.19**.
"P(X ≤ x) = f(x) = pdf"; "cumulative density function"	P(X ≤ x) = F(x) is the cdf, the cumulative **distribution** function.
Exponential: "λ = rata-rata"; P(X = 88) = f(88)	mean = 1/λ; P(X = 88) = 0 for a continuous X.
56 · 0.64 · 0.077 for B(8, 0.4); σ = 0.913 for variance 0.84	0.4^3 = 0.064; √0.84 = 0.9165.
"s = ±2", "$\sigma^2 > 0$"	s = +√s^2 only; variance ≥ 0.

General traps

- Discrete: P(X < 3) = P(X ≤ 2). Continuous: ≤ and < are the same.
- Without replacement from a small lot → Hypergeometric, not Binomial.
- Independent ≠ mutually exclusive.
- P(A | B) ≠ P(B | A). Use Bayes to flip.
- Var(aX + b) = a^2Var(X).
- Exponential: λ = 1/mean.
- Round z to 2 decimals for the table only.
- Indonesian numbers: 0,4 = 0.4 and 15.015 = 15015.

T.1 Standard Normal Table: Φ(z) = P(Z ≤ z), z ≤ 0

Row = z to one decimal, column = second decimal. For negative z: z = -1.36 → row -1.3, column .06. Area is to the **LEFT** of z *(luas di kiri z)*.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

T.2 Standard Normal Table: Φ(z) = P(Z ≤ z), z ≥ 0

Row = z to one decimal, column = second decimal. z = 1.36 → row 1.3, column .06 → .9131. Area is to the **LEFT** of z *(luas di kiri z)*.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Handy critical values: $z_{.10} = 1.28$, $z_{.05} = 1.645$, $z_{.025} = 1.96$, $z_{.01} = 2.33$, $z_{.005} = 2.58$ (area to the RIGHT). Beyond the table: z ≥ 3.50 → Φ ≈ 1.0000; z ≤ -3.50 → Φ ≈ 0.0000.

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x)

P(X = k) = B(k) - B(k-1). P(X ≥ k) = 1 - B(k-1). P(X > k) = 1 - B(k). Calculator check: **Binomial CD** gives the same value.

n = 5 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.7738	.5905	.3277	.2373	.1681	.0778	.0312	.0102	.0024	.0010	.0003	.0000	.0000
1	.9774	.9185	.7373	.6328	.5282	.3370	.1875	.0870	.0308	.0156	.0067	.0005	.0000
2	.9988	.9914	.9421	.8965	.8369	.6826	.5000	.3174	.1631	.1035	.0579	.0086	.0012
3	1.000	.9995	.9933	.9844	.9692	.9130	.8125	.6630	.4718	.3672	.2627	.0815	.0226
4	1.000	1.000	.9997	.9990	.9976	.9898	.9688	.9222	.8319	.7627	.6723	.4095	.2262
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 6 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.7351	.5314	.2621	.1780	.1176	.0467	.0156	.0041	.0007	.0002	.0001	.0000	.0000
1	.9672	.8857	.6554	.5339	.4202	.2333	.1094	.0410	.0109	.0046	.0016	.0001	.0000
2	.9978	.9842	.9011	.8306	.7443	.5443	.3438	.1792	.0705	.0376	.0170	.0013	.0001
3	.9999	.9987	.9830	.9624	.9295	.8208	.6562	.4557	.2557	.1694	.0989	.0158	.0022
4	1.000	.9999	.9984	.9954	.9891	.9590	.8906	.7667	.5798	.4661	.3446	.1143	.0328
5	1.000	1.000	.9999	.9998	.9993	.9959	.9844	.9533	.8824	.8220	.7379	.4686	.2649
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 7 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6983	.4783	.2097	.1335	.0824	.0280	.0078	.0016	.0002	.0001	.0000	.0000	.0000
1	.9556	.8503	.5767	.4449	.3294	.1586	.0625	.0188	.0038	.0013	.0004	.0000	.0000
2	.9962	.9743	.8520	.7564	.6471	.4199	.2266	.0963	.0288	.0129	.0047	.0002	.0000
3	.9998	.9973	.9667	.9294	.8740	.7102	.5000	.2898	.1260	.0706	.0333	.0027	.0002
4	1.000	.9998	.9953	.9871	.9712	.9037	.7734	.5801	.3529	.2436	.1480	.0257	.0038
5	1.000	1.000	.9996	.9987	.9962	.9812	.9375	.8414	.6706	.5551	.4233	.1497	.0444
6	1.000	1.000	1.000	.9999	.9998	.9984	.9922	.9720	.9176	.8665	.7903	.5217	.3017
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 8 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6634	.4305	.1678	.1001	.0576	.0168	.0039	.0007	.0001	.0000	.0000	.0000	.0000
1	.9428	.8131	.5033	.3671	.2553	.1064	.0352	.0085	.0013	.0004	.0001	.0000	.0000
2	.9942	.9619	.7969	.6785	.5518	.3154	.1445	.0498	.0113	.0042	.0012	.0000	.0000
3	.9996	.9950	.9437	.8862	.8059	.5941	.3633	.1737	.0580	.0273	.0104	.0004	.0000
4	1.000	.9996	.9896	.9727	.9420	.8263	.6367	.4059	.1941	.1138	.0563	.0050	.0004
5	1.000	1.000	.9988	.9958	.9887	.9502	.8555	.6846	.4482	.3215	.2031	.0381	.0058
6	1.000	1.000	.9999	.9996	.9987	.9915	.9648	.8936	.7447	.6329	.4967	.1869	.0572
7	1.000	1.000	1.000	1.000	.9999	.9993	.9961	.9832	.9424	.8999	.8322	.5695	.3366
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 9 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6302	.3874	.1342	.0751	.0404	.0101	.0020	.0003	.0000	.0000	.0000	.0000	.0000
1	.9288	.7748	.4362	.3003	.1960	.0705	.0195	.0038	.0004	.0001	.0000	.0000	.0000
2	.9916	.9470	.7382	.6007	.4628	.2318	.0898	.0250	.0043	.0013	.0003	.0000	.0000
3	.9994	.9917	.9144	.8343	.7297	.4826	.2539	.0994	.0253	.0100	.0031	.0001	.0000
4	1.000	.9991	.9804	.9511	.9012	.7334	.5000	.2666	.0988	.0489	.0196	.0009	.0000
5	1.000	.9999	.9969	.9900	.9747	.9006	.7461	.5174	.2703	.1657	.0856	.0083	.0006
6	1.000	1.000	.9997	.9987	.9957	.9750	.9102	.7682	.5372	.3993	.2618	.0530	.0084
7	1.000	1.000	1.000	.9999	.9996	.9962	.9805	.9295	.8040	.6997	.5638	.2252	.0712
8	1.000	1.000	1.000	1.000	1.000	.9997	.9980	.9899	.9596	.9249	.8658	.6126	.3698
9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x) (cont.)

n = 10 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.5987	.3487	.1074	.0563	.0282	.0060	.0010	.0001	.0000	.0000	.0000	.0000	.0000
1	.9139	.7361	.3758	.2440	.1493	.0464	.0107	.0017	.0001	.0000	.0000	.0000	.0000
2	.9885	.9298	.6778	.5256	.3828	.1673	.0547	.0123	.0016	.0004	.0001	.0000	.0000
3	.9990	.9872	.8791	.7759	.6496	.3823	.1719	.0548	.0106	.0035	.0009	.0000	.0000
4	.9999	.9984	.9672	.9219	.8497	.6331	.3770	.1662	.0473	.0197	.0064	.0001	.0000
5	1.000	.9999	.9936	.9803	.9527	.8338	.6230	.3669	.1503	.0781	.0328	.0016	.0001
6	1.000	1.000	.9991	.9965	.9894	.9452	.8281	.6177	.3504	.2241	.1209	.0128	.0010
7	1.000	1.000	.9999	.9996	.9984	.9877	.9453	.8327	.6172	.4744	.3222	.0702	.0115
8	1.000	1.000	1.000	1.000	.9999	.9983	.9893	.9536	.8507	.7560	.6242	.2639	.0861
9	1.000	1.000	1.000	1.000	1.000	.9999	.9990	.9940	.9718	.9437	.8926	.6513	.4013
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 12 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.5404	.2824	.0687	.0317	.0138	.0022	.0002	.0000	.0000	.0000	.0000	.0000	.0000
1	.8816	.6590	.2749	.1584	.0850	.0196	.0032	.0003	.0000	.0000	.0000	.0000	.0000
2	.9804	.8891	.5583	.3907	.2528	.0834	.0193	.0028	.0002	.0000	.0000	.0000	.0000
3	.9978	.9744	.7946	.6488	.4925	.2253	.0730	.0153	.0017	.0004	.0001	.0000	.0000
4	.9998	.9957	.9274	.8424	.7237	.4382	.1938	.0573	.0095	.0028	.0006	.0000	.0000
5	1.000	.9995	.9806	.9456	.8822	.6652	.3872	.1582	.0386	.0143	.0039	.0001	.0000
6	1.000	.9999	.9961	.9857	.9614	.8418	.6128	.3348	.1178	.0544	.0194	.0005	.0000
7	1.000	1.000	.9994	.9972	.9905	.9427	.8062	.5618	.2763	.1576	.0726	.0043	.0002
8	1.000	1.000	.9999	.9996	.9983	.9847	.9270	.7747	.5075	.3512	.2054	.0256	.0022
9	1.000	1.000	1.000	1.000	.9998	.9972	.9807	.9166	.7472	.6093	.4417	.1109	.0196
10	1.000	1.000	1.000	1.000	1.000	.9997	.9968	.9804	.9150	.8416	.7251	.3410	.1184
11	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9978	.9862	.9683	.9313	.7176	.4596
12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 15 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.4633	.2059	.0352	.0134	.0047	.0005	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.8290	.5490	.1671	.0802	.0353	.0052	.0005	.0000	.0000	.0000	.0000	.0000	.0000
2	.9638	.8159	.3980	.2361	.1268	.0271	.0037	.0003	.0000	.0000	.0000	.0000	.0000
3	.9945	.9444	.6482	.4613	.2969	.0905	.0176	.0019	.0001	.0000	.0000	.0000	.0000
4	.9994	.9873	.8358	.6865	.5155	.2173	.0592	.0093	.0007	.0001	.0000	.0000	.0000
5	.9999	.9978	.9389	.8516	.7216	.4032	.1509	.0338	.0037	.0008	.0001	.0000	.0000
6	1.000	.9997	.9819	.9434	.8689	.6098	.3036	.0950	.0152	.0042	.0008	.0000	.0000
7	1.000	1.000	.9958	.9827	.9500	.7869	.5000	.2131	.0500	.0173	.0042	.0000	.0000
8	1.000	1.000	.9992	.9958	.9848	.9050	.6964	.3902	.1311	.0566	.0181	.0003	.0000
9	1.000	1.000	.9999	.9992	.9963	.9662	.8491	.5968	.2784	.1484	.0611	.0022	.0001
10	1.000	1.000	1.000	.9999	.9993	.9907	.9408	.7827	.4845	.3135	.1642	.0127	.0006
11	1.000	1.000	1.000	1.000	.9999	.9981	.9824	.9095	.7031	.5387	.3518	.0556	.0055
12	1.000	1.000	1.000	1.000	1.000	.9997	.9963	.9729	.8732	.7639	.6020	.1841	.0362
13	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9948	.9647	.9198	.8329	.4510	.1710
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9953	.9866	.9648	.7941	.5367
15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x) (cont.)

n = 20 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.3585	.1216	.0115	.0032	.0008	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.7358	.3917	.0692	.0243	.0076	.0005	.0000	.0000	.0000	.0000	.0000	.0000	.0000
2	.9245	.6769	.2061	.0913	.0355	.0036	.0002	.0000	.0000	.0000	.0000	.0000	.0000
3	.9841	.8670	.4114	.2252	.1071	.0160	.0013	.0000	.0000	.0000	.0000	.0000	.0000
4	.9974	.9568	.6296	.4148	.2375	.0510	.0059	.0003	.0000	.0000	.0000	.0000	.0000
5	.9997	.9887	.8042	.6172	.4164	.1256	.0207	.0016	.0000	.0000	.0000	.0000	.0000
6	1.000	.9976	.9133	.7858	.6080	.2500	.0577	.0065	.0003	.0000	.0000	.0000	.0000
7	1.000	.9996	.9679	.8982	.7723	.4159	.1316	.0210	.0013	.0002	.0000	.0000	.0000
8	1.000	.9999	.9900	.9591	.8867	.5956	.2517	.0565	.0051	.0009	.0001	.0000	.0000
9	1.000	1.000	.9974	.9861	.9520	.7553	.4119	.1275	.0171	.0039	.0006	.0000	.0000
10	1.000	1.000	.9994	.9961	.9829	.8725	.5881	.2447	.0480	.0139	.0026	.0000	.0000
11	1.000	1.000	.9999	.9991	.9949	.9435	.7483	.4044	.1133	.0409	.0100	.0001	.0000
12	1.000	1.000	1.000	.9998	.9987	.9790	.8684	.5841	.2277	.1018	.0321	.0004	.0000
13	1.000	1.000	1.000	1.000	.9997	.9935	.9423	.7500	.3920	.2142	.0867	.0024	.0000
14	1.000	1.000	1.000	1.000	1.000	.9984	.9793	.8744	.5836	.3828	.1958	.0113	.0003
15	1.000	1.000	1.000	1.000	1.000	.9997	.9941	.9490	.7625	.5852	.3704	.0432	.0026
16	1.000	1.000	1.000	1.000	1.000	1.000	.9987	.9840	.8929	.7748	.5886	.1330	.0159
17	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9964	.9645	.9087	.7939	.3231	.0755
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9924	.9757	.9308	.6083	.2642
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9992	.9968	.9885	.8784	.6415
20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 25 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.2774	.0718	.0038	.0008	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.6424	.2712	.0274	.0070	.0016	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000
2	.8729	.5371	.0982	.0321	.0090	.0004	.0000	.0000	.0000	.0000	.0000	.0000	.0000
3	.9659	.7636	.2340	.0962	.0332	.0024	.0001	.0000	.0000	.0000	.0000	.0000	.0000
4	.9928	.9020	.4207	.2137	.0905	.0095	.0005	.0000	.0000	.0000	.0000	.0000	.0000
5	.9988	.9666	.6167	.3783	.1935	.0294	.0020	.0001	.0000	.0000	.0000	.0000	.0000
6	.9998	.9905	.7800	.5611	.3407	.0736	.0073	.0003	.0000	.0000	.0000	.0000	.0000
7	1.000	.9977	.8909	.7265	.5118	.1536	.0216	.0012	.0000	.0000	.0000	.0000	.0000
8	1.000	.9995	.9532	.8506	.6769	.2735	.0539	.0043	.0001	.0000	.0000	.0000	.0000
9	1.000	.9999	.9827	.9287	.8106	.4246	.1148	.0132	.0005	.0000	.0000	.0000	.0000
10	1.000	1.000	.9944	.9703	.9022	.5858	.2122	.0344	.0018	.0002	.0000	.0000	.0000
11	1.000	1.000	.9985	.9893	.9558	.7323	.3450	.0778	.0060	.0009	.0001	.0000	.0000
12	1.000	1.000	.9996	.9966	.9825	.8462	.5000	.1538	.0175	.0034	.0004	.0000	.0000
13	1.000	1.000	.9999	.9991	.9940	.9222	.6550	.2677	.0442	.0107	.0015	.0000	.0000
14	1.000	1.000	1.000	.9998	.9982	.9656	.7878	.4142	.0978	.0297	.0056	.0000	.0000
15	1.000	1.000	1.000	1.000	.9995	.9868	.8852	.5754	.1894	.0713	.0173	.0001	.0000
16	1.000	1.000	1.000	1.000	.9999	.9957	.9461	.7265	.3231	.1494	.0468	.0005	.0000
17	1.000	1.000	1.000	1.000	1.000	.9988	.9784	.8464	.4882	.2735	.1091	.0023	.0000
18	1.000	1.000	1.000	1.000	1.000	.9997	.9927	.9264	.6593	.4389	.2200	.0095	.0002
19	1.000	1.000	1.000	1.000	1.000	.9999	.9980	.9706	.8065	.6217	.3833	.0334	.0012
20	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9905	.9095	.7863	.5793	.0980	.0072
21	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9976	.9668	.9038	.7660	.2364	.0341
22	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9996	.9910	.9679	.9018	.4629	.1271
23	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9984	.9930	.9726	.7288	.3576
24	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9992	.9962	.9282	.7226
25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.4 Cumulative Poisson: F(x; λ) = P(X ≤ x)

λ = mean number of events in the interval (scale it to the interval asked: λ_{new} = rate × length). P(X = k) = F(k) - F(k-1). Calculator check: **Poisson CD**.

x \ λ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0	.9048	.8187	.7408	.6703	.6065	.5488	.4966	.4493	.4066	.3679
1	.9953	.9825	.9631	.9384	.9098	.8781	.8442	.8088	.7725	.7358
2	.9998	.9989	.9964	.9921	.9856	.9769	.9659	.9526	.9371	.9197
3	1.000	.9999	.9997	.9992	.9982	.9966	.9942	.9909	.9865	.9810
4	1.000	1.000	1.000	.9999	.9998	.9996	.9992	.9986	.9977	.9963
5	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9997	.9994
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

x \ λ	1.5	2	2.5	3	3.5	4	4.5	5	6	7
0	.2231	.1353	.0821	.0498	.0302	.0183	.0111	.0067	.0025	.0009
1	.5578	.4060	.2873	.1991	.1359	.0916	.0611	.0404	.0174	.0073
2	.8088	.6767	.5438	.4232	.3208	.2381	.1736	.1247	.0620	.0296
3	.9344	.8571	.7576	.6472	.5366	.4335	.3423	.2650	.1512	.0818
4	.9814	.9473	.8912	.8153	.7254	.6288	.5321	.4405	.2851	.1730
5	.9955	.9834	.9580	.9161	.8576	.7851	.7029	.6160	.4457	.3007
6	.9991	.9955	.9858	.9665	.9347	.8893	.8311	.7622	.6063	.4497
7	.9998	.9989	.9958	.9881	.9733	.9489	.9134	.8666	.7440	.5987
8	1.000	.9998	.9989	.9962	.9901	.9786	.9597	.9319	.8472	.7291
9	1.000	1.000	.9997	.9989	.9967	.9919	.9829	.9682	.9161	.8305
10	1.000	1.000	.9999	.9997	.9990	.9972	.9933	.9863	.9574	.9015
11	1.000	1.000	1.000	.9999	.9997	.9991	.9976	.9945	.9799	.9467
12	1.000	1.000	1.000	1.000	.9999	.9997	.9992	.9980	.9912	.9730
13	1.000	1.000	1.000	1.000	1.000	.9999	.9997	.9993	.9964	.9872
14	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9986	.9943
15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9995	.9976
16	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9990
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999

T.4 Cumulative Poisson (cont.)

x \ λ	8	9	10	11	12	13	14	15	18	20
0	.0003	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.0030	.0012	.0005	.0002	.0001	.0000	.0000	.0000	.0000	.0000
2	.0138	.0062	.0028	.0012	.0005	.0002	.0001	.0000	.0000	.0000
3	.0424	.0212	.0103	.0049	.0023	.0011	.0005	.0002	.0000	.0000
4	.0996	.0550	.0293	.0151	.0076	.0037	.0018	.0009	.0001	.0000
5	.1912	.1157	.0671	.0375	.0203	.0107	.0055	.0028	.0003	.0001
6	.3134	.2068	.1301	.0786	.0458	.0259	.0142	.0076	.0010	.0003
7	.4530	.3239	.2202	.1432	.0895	.0540	.0316	.0180	.0029	.0008
8	.5925	.4557	.3328	.2320	.1550	.0998	.0621	.0374	.0071	.0021
9	.7166	.5874	.4579	.3405	.2424	.1658	.1094	.0699	.0154	.0050
10	.8159	.7060	.5830	.4599	.3472	.2517	.1757	.1185	.0304	.0108
11	.8881	.8030	.6968	.5793	.4616	.3532	.2600	.1848	.0549	.0214
12	.9362	.8758	.7916	.6887	.5760	.4631	.3585	.2676	.0917	.0390
13	.9658	.9261	.8645	.7813	.6815	.5730	.4644	.3632	.1426	.0661
14	.9827	.9585	.9165	.8540	.7720	.6751	.5704	.4657	.2081	.1049
15	.9918	.9780	.9513	.9074	.8444	.7636	.6694	.5681	.2867	.1565
16	.9963	.9889	.9730	.9441	.8987	.8355	.7559	.6641	.3751	.2211
17	.9984	.9947	.9857	.9678	.9370	.8905	.8272	.7489	.4686	.2970
18	.9993	.9976	.9928	.9823	.9626	.9302	.8826	.8195	.5622	.3814
19	.9997	.9989	.9965	.9907	.9787	.9573	.9235	.8752	.6509	.4703
20	.9999	.9996	.9984	.9953	.9884	.9750	.9521	.9170	.7307	.5591
21	1.000	.9998	.9993	.9977	.9939	.9859	.9712	.9469	.7991	.6437
22	1.000	.9999	.9997	.9990	.9970	.9924	.9833	.9673	.8551	.7206
23	1.000	1.000	.9999	.9995	.9985	.9960	.9907	.9805	.8989	.7875
24	1.000	1.000	1.000	.9998	.9993	.9980	.9950	.9888	.9317	.8432
25	1.000	1.000	1.000	.9999	.9997	.9990	.9974	.9938	.9554	.8878
26	1.000	1.000	1.000	1.000	.9999	.9995	.9987	.9967	.9718	.9221
27	1.000	1.000	1.000	1.000	.9999	.9998	.9994	.9983	.9827	.9475
28	1.000	1.000	1.000	1.000	1.000	.9999	.9997	.9991	.9897	.9657
29	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996	.9941	.9782
30	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9967	.9865
31	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9982	.9919
32	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9990	.9953
33	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9973
34	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9985
35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9992
36	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996

R · Real Quiz 1 (LA40 + LB40), fully solved

These are the prof's actual questions. Questions 5 and 6 were identical in both classes. Each part is laid out as **Known → Method → Working → Answer → Check**, which is a good layout to copy onto your answer sheet.

LA40 · Question 1: variable type and measurement scale

Soal. Laboratorium Computer Engineering meneliti kinerja perangkat dengan variabel: jenis sistem kontrol, waktu respons sensor, jenis sensor, kecepatan pemrosesan data, jenis mikrokontroler, tingkat akurasi sensor, jumlah perangkat yang terhubung, status operasional mesin, konsumsi daya sistem, jenis protokol komunikasi, waktu siklus proses otomatis, suhu operasional perangkat, mode pengoperasian sistem, tingkat kesalahan pembacaan sensor, jumlah kegagalan sistem otomatis. Sebutkan jenis variabel dan skala pengukuran masing-masing.

For each of the 15 variables, give the variable type and the measurement scale.

Method: the decision steps on **p.8**. Rule of thumb: "jenis ..." (type of) is nominal, "waktu / kecepatan / daya / jumlah" is ratio, "suhu °C" is interval.

#	Variabel	Jenis (type)	Skala	Alasan (reason)
1	Jenis sistem kontrol	Kualitatif	Nominal	categories (open-loop, closed-loop, PID) with no order
2	Waktu respons sensor	Kuantitatif kontinu	Rasio	measured time; 0 ms = no time
3	Jenis sensor	Kualitatif	Nominal	labels (ultrasonik, IR, suhu)
4	Kecepatan pemrosesan data	Kuantitatif kontinu	Rasio	e.g. MB/s; 0 = no processing
5	Jenis mikrokontroler	Kualitatif	Nominal	labels (Arduino, ESP32, STM32)
6	Tingkat akurasi sensor	Kuantitatif kontinu	Rasio	a percentage; 0% = no accuracy. *If given as tinggi/sedang/rendah: kualitatif, ordinal.*
7	Jumlah perangkat yang terhubung	Kuantitatif diskrit	Rasio	a count; 0 = no devices
8	Status operasional mesin	Kualitatif	Nominal	ON/OFF/standby, no order
9	Konsumsi daya sistem	Kuantitatif kontinu	Rasio	watts; 0 W = no power
10	Jenis protokol komunikasi	Kualitatif	Nominal	labels (I2C, SPI, UART, Wi-Fi)
11	Waktu siklus proses otomatis	Kuantitatif kontinu	Rasio	measured time
12	Suhu operasional perangkat	Kuantitatif kontinu	**Interval**	°C: 0°C is not "no heat", and 40°C is not "twice as hot" as 20°C
13	Mode pengoperasian sistem	Kualitatif	Nominal	manual/otomatis. *If the modes are levels (manual < semi-otomatis < otomatis), ordinal is defensible. Write the assumption.*
14	Tingkat kesalahan pembacaan sensor	Kuantitatif kontinu	Rasio	error rate in %; 0% = no errors
15	Jumlah kegagalan sistem otomatis	Kuantitatif diskrit	Rasio	a count; 0 = no failures

Summary line for the answer sheet: Kualitatif nominal: 1, 3, 5, 8, 10, 13. Kuantitatif kontinu rasio: 2, 4, 6, 9, 11, 14. Kuantitatif diskrit rasio: 7, 15. Kuantitatif kontinu interval: 12.

LB40 · Question 3: choose 3 numeric and 3 categorical variables, with scales

Soal. Seorang engineer mengembangkan sensor otomatis untuk inspeksi kualitas barang. Tentukan 3 variabel numerik dan 3 variabel kategorik, beserta skala pengukurannya.

Propose 3 numerical and 3 categorical variables for an automatic quality-inspection sensor, and give each one's scale.

Any sensible answer works. The marks come from the correct scale and a reason. A strong answer covers all four scales.

	Variabel	Jenis	Skala	Alasan
Numerik	Berat barang (gram)	Kuantitatif kontinu	Rasio	0 g = no weight; 200 g is twice 100 g
	Jumlah cacat per barang	Kuantitatif diskrit	Rasio	a count, true zero
	Suhu permukaan barang (°C)	Kuantitatif kontinu	Interval	0°C is not "no temperature"
Kategorik	Status inspeksi (lolos / tidak lolos)	Kualitatif	Nominal	two labels, no order
	Jenis cacat (gores, retak, penyok)	Kualitatif	Nominal	labels, no order
	Grade kualitas (A / B / C)	Kualitatif	Ordinal	A > B > C, but the gaps are not measurable

Other good choices: dimensi/panjang (rasio), waktu inspeksi (rasio), kecerahan warna (rasio), warna barang (nominal), tingkat keparahan cacat ringan/sedang/berat (ordinal).

LA40 · Question 2: signal amplitude data (n = 15, mV)

Soal. Data amplitudo sinyal sensor getaran (mV): 22, 75, 24, 67, 23, 26, 25, 28, 24, 26, 23, 25, 29, 27, 65.

a. Hitunglah nilai mean
b. Tentukan five number summary
c. Hitung coefficient of variation
d. Buatlah diagram stem and leaf
e. Gambarkan boxplot dan identifikasi apakah terdapat outlier.

Step 0: sort and number the data (do this first; the bold values are Q1, median, Q3)

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value	22	23	23	**24**	24	25	25	**26**	26	27	28	**29**	65	67	75

n = 15 · Σx = 509 · Σx^2 = 21989

a. Mean

METHOD $\bar{x} = \frac{\sum x_i}{n}$

WORKING $\bar{x} = \frac{509}{15}$

ANSWER **x̄ = 33.9333 mV**

CHECK STAT 1-VAR → x̄ = 33.9333. The mean is far above most values (22–29) because of 65, 67, 75. That already hints at outliers.

b. Five-number summary

METHOD Positions for n = 15: Q1 at 0.25(16) = **4**, median at 0.5(16) = **8**, Q3 at 0.75(16) = **12**.

WORKING Min = $x_{(1)}$ = 22 · Q1 = $x_{(4)}$ = 24 · Median = $x_{(8)}$ = 26 · Q3 = $x_{(12)}$ = 29 · Max = $x_{(15)}$ = 75

ANSWER **Min 22, Q1 24, Median 26, Q3 29, Max 75**

c. Coefficient of variation

METHOD $s^2 = \frac{\sum x^2 - (\sum x)^2/n}{n-1}$, then $CV = \frac{s}{\bar{x}} \times 100\%$

WORKING

$$s^2 = \frac{21989 - 509^2/15}{14} = \frac{21989 - 17272.0667}{14} = \frac{4716.9333}{14} = 336.9238$$

$$s = \sqrt{336.9238} = 18.3555 \text{ mV}$$

$$CV = \frac{18.3555}{33.9333} \times 100\%$$

ANSWER **CV = 54.09%** . The amplitude is **not stable**: the SD is more than half of the mean, mainly because of the three large values.

CHECK STAT → sx = 18.3555. With σx (÷ n) you would get 52.26%, which is the wrong formula for a sample.

d. Stem-and-leaf

METHOD stem = tens digit, leaf = ones digit. Keep the empty stems 3, 4, 5 to show the gap.

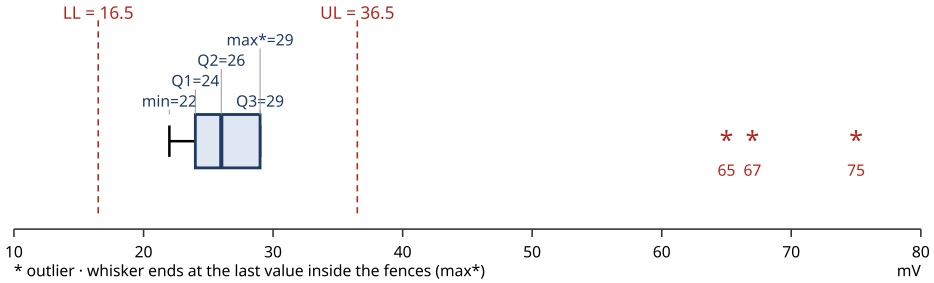
ANSWER

```
Stem | Leaf
   2 | 2 3 3 4 4 5 5 6 6 7 8 9
   3 |
   4 |
   5 |
   6 | 5 7
   7 | 5
Key: 2 | 2 = 22 mV
```

e. Boxplot and outliers

WORKING IQR = 29 - 24 = 5
LL = 24 - 1.5(5) = 16.5
UL = 29 + 1.5(5) = 36.5
65, 67, 75 > 36.5. The largest value inside is 29, so the upper whisker ends at 29. The lowest value 22 > 16.5, so the lower whisker ends at 22.

ANSWER **Outliers: 65, 67, 75 mV**. The rest of the data is tightly packed and right-skewed (mean 33.9 > median 26).



LB40 · Question 1: machine temperature data (n = 15, °C)

Soal. Data suhu mesin (°C): 85, 16, 24, 27, 25, 56, 28, 24, 25, 27, 16, 25, 53, 76, 58.

a. mean
b. median
c. interquartile range (IQR)
d. standar deviasi
e. Gambarkan boxplotnya. Tentukan apakah terdapat outlier.

Step 0: sort (bold = Q1, median, Q3)

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value	16	16	24	**24**	25	25	25	**27**	27	28	53	**56**	58	76	85

n = 15 · Σx = 565 · Σx^2 = 28091

a. Mean

WORKING $\bar{x} = \frac{565}{15}$

ANSWER **x̄ = 37.6667 °C**

b. Median

WORKING position 0.5(15 + 1) = 8 → $x_{(8)}$

ANSWER **Median = 27 °C**. The median is below the mean, so the data are skewed right.

c. IQR

WORKING Q1 = $x_{(4)}$ = 24 · Q3 = $x_{(12)}$ = 56 · IQR = 56 - 24

ANSWER **IQR = 32 °C**

d. Standard deviation (sample)

WORKING

$$s^2 = \frac{28091 - 565^2/15}{14} = \frac{28091 - 21281.6667}{14} = \frac{6809.3333}{14} = 486.3810$$

$$s = \sqrt{486.3810}$$

ANSWER **s = 22.0540 °C**

CHECK STAT → sx = 22.0540 (σx = 21.3062 is the population version). CV would be 22.054/37.667 = 58.55%.

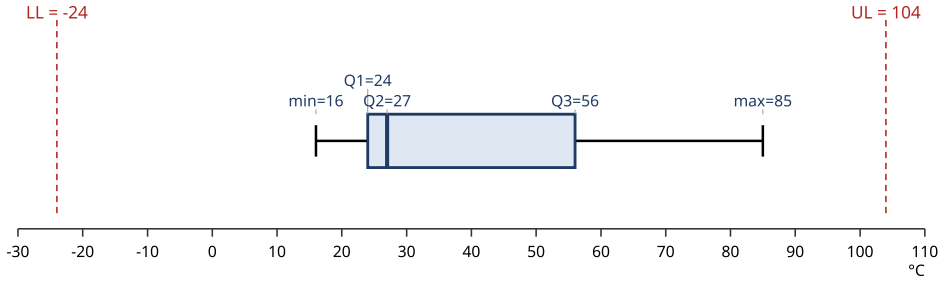
e. Boxplot and outliers

WORKING LL = 24 - 1.5(32) = -24 · UL = 56 + 1.5(32) = 104. Every value is between -24 and 104, so the whiskers run to the min 16 and the max 85.

ANSWER **No outliers** *(tidak terdapat outlier)*. The box is long on the right (median 27 is close to Q1 = 24), so the distribution is skewed right.

TRAP 76 and 85 *look* extreme, but IQR = 32 makes the fences wide. Compute the fences; never guess.

Five-number summary (not asked, but useful for drawing): 16, 24, 27, 56, 85. Stem-and-leaf: `1|6 6 · 2|4 4 5 5 5 7 7 8 · 3| · 4| · 5|3 6 8 · 6| · 7|6 · 8|5`



LA40 · Question 3: contingency table (OS × condition)

Step 0: add the totals

Sistem Operasi	Baik	Sedang	Rusak	Total
Windows	40	10	10	**60**
Linux	15	15	10	**40**
macOS	20	5	5	**30**
Total	**75**	**30**	**25**	**130**

a. P(macOS dan sedang): joint

METHOD cell ÷ grand total

ANSWER $\mathrm{P}(M \cap S) = \frac{5}{130} = \mathbf{0.0385}$

b. P(Windows atau rusak): union

METHOD $\mathrm{P}(W \cup R) = \mathrm{P}(W) + \mathrm{P}(R) - \mathrm{P}(W \cap R)$

WORKING $\frac{60}{130} + \frac{25}{130} - \frac{10}{130} = \frac{75}{130}$

ANSWER **0.5769**

CHECK Count the cells directly: Windows row 40 + 10 + 10 plus the other rusak cells 10 + 5 = 75 ✓

c. P(Linux | rusak): conditional

METHOD The condition is "rusak", so divide by the rusak column total (25).

ANSWER $\mathrm{P}(L \mid R) = \frac{10}{25} = \mathbf{0.4}$

d. P(sedang | macOS)

ANSWER $\frac{5}{30} = \mathbf{0.1667}$ (divide by the macOS row total)

e. P(Windows | baik)

ANSWER $\frac{40}{75} = \mathbf{0.5333}$ (divide by the baik column total)

f. Are OS type and condition independent?

METHOD Independent only if $\mathrm{P}(A \mid B) = \mathrm{P}(A)$ for **every** cell. One counter-example is enough to say "no".

WORKING $\mathrm{P}(W \mid \text{Baik}) = 0.5333$ but $\mathrm{P}(W) = \frac{60}{130} = 0.4615$.
Or: $\mathrm{P}(W \cap \text{Baik}) = \frac{40}{130} = 0.3077$ vs $\mathrm{P}(W)\,\mathrm{P}(\text{Baik}) = 0.4615 \times \frac{75}{130} = 0.2663$.

ANSWER **Not independent** *(tidak saling bebas)*: Windows computers are more often in good condition than computers overall.

LB40 · Question 2: contingency table (CPU temperature × system failure)

Suhu CPU	Gagal (G)	Tidak Gagal (TG)	Total
Normal (N)	10	110	**120**
Tinggi (T)	40	40	**80**
Total	**50**	**150**	**200**

a. P(tinggi dan gagal)

ANSWER 40/200 = **0.20**

b. P(normal dan tidak gagal)

ANSWER 110/200 = **0.55**

c. P(gagal): marginal

ANSWER 50/200 = **0.25**

d. P(tinggi atau gagal)

WORKING (80 + 50 - 40)/200 = 90/200

ANSWER **0.45**

e. P(tinggi | gagal)

WORKING divide by the gagal total: 40/50

ANSWER **0.80**

f. P(tidak gagal | normal)

WORKING divide by the normal total: 110/120

ANSWER **0.9167**

✔ Not asked, but expect it on the midterm: are temperature and failure independent? P(G | T) = 40/80 = 0.5, but P(G) = 0.25, so **no**. A hot CPU doubles the failure rate: hot CPUs are 40% of the computers but 80% of the failures.

LA40 · Question 4: Binomial, RAM modules (8% fail, n = 20)

Known: X = banyaknya modul RAM gagal. Each module fails independently with p = 0.08 and n = 20 are tested, so **X ~ B(20, 0.08)**.
$P(X = x) = \binom{20}{x}(0.08)^x(0.92)^{20-x}$ → P(0) = 0.1887, P(1) = 0.3282, P(2) = 0.2711, P(3) = 0.1414, P(4) = 0.0523, P(5) = 0.0145

a. Paling banyak 2: P(X ≤ 2)

WORKING P(0) + P(1) + P(2) = 0.1887 + 0.3282 + 0.2711
e.g. $P(2) = \binom{20}{2}(0.08)^2(0.92)^{18} = 190(0.0064)(0.2229) = 0.2711$

ANSWER **0.7879**

CHECK Binomial CD: x = 2, N = 20, p = 0.08 → 0.78795

b. Setidaknya 4: P(X ≥ 4)

WORKING 1 - P(X ≤ **3**) = 1 - (0.1887 + 0.3282 + 0.2711 + 0.1414) = 1 - 0.9294

ANSWER **0.0706**

TRAP 1 - CD(4) = 0.0183 is wrong: it drops X = 4.

c. Dari 2 hingga 5: P(2 ≤ X ≤ 5)

WORKING P(2) + P(3) + P(4) + P(5) = 0.2711 + 0.1414 + 0.0523 + 0.0145 (= CD(5) - CD(1))

ANSWER **0.4793**

d. Mean and SD

WORKING μ = np = 20(0.08) · σ^2 = np(1 - p) = 20(0.08)(0.92) = 1.472 · σ = √1.472

ANSWER **μ = 1.6 modules, σ = 1.2133**

LB40 · Question 4: Binomial, data packets (8% error, n = 12)

Known: X = banyaknya paket error, **X ~ B(12, 0.08)**. P(0) = 0.92^{12} = 0.3677, P(1) = 0.3837, P(2) = 0.1835, P(3) = 0.0532.

a. Tepat 3: P(X = 3)

WORKING $\binom{12}{3}(0.08)^3(0.92)^9 = 220(0.000512)(0.4722)$

ANSWER **0.0532** (Binomial PD → 0.05318)

b. Paling banyak 2: P(X ≤ 2)

WORKING 0.3677 + 0.3837 + 0.1835

ANSWER **0.9348** (CD → 0.93480; adding rounded terms gives 0.9349)

c. Setidaknya 1: P(X ≥ 1)

WORKING 1 - P(X = 0) = 1 - 0.92^{12} = 1 - 0.3677

ANSWER **0.6323**

d. Mean and SD

WORKING μ = 12(0.08) · σ = √(12 · 0.08 · 0.92) = √0.8832

ANSWER **μ = 0.96 packets, σ = 0.9398**

LA40 + LB40 · Question 5: Poisson, server errors (4 per hour)

Known: X = banyaknya system error per jam, **X ~ Poisson(λ = 4)**. $\mathrm{P}(X = x) = \frac{e^{-4}4^x}{x!}$, with e^{-4} = 0.018316. P(0) = 0.0183, P(1) = 0.0733, P(2) = 0.1465, P(3) = 0.1954. Parts a–c are per hour, because no other interval is stated.

a. Tepat 3: P(X = 3)

WORKING $\frac{e^{-4}4^3}{3!} = \frac{0.018316 \times 64}{6}$

ANSWER **0.1954** (Poisson PD x = 3, μ = 4)

b. Paling banyak 2: P(X ≤ 2)

WORKING $e^{-4}\left(1 + 4 + \frac{16}{2}\right) = 13e^{-4}$

ANSWER **0.2381**

c. Setidaknya 1: P(X ≥ 1)

WORKING 1 - P(0) = 1 - e^{-4}

ANSWER **0.9817**

d. Tidak ada error selama 30 menit

METHOD **Rescale λ**: 4 per 60 minutes → λ = 4 × 30/60 = 2 per 30 minutes. Y ~ Poisson(2).

WORKING $\mathrm{P}(Y = 0) = \frac{e^{-2}2^0}{0!} = e^{-2}$

ANSWER **0.1353**

TRAP Using λ = 4 gives 0.0183, the probability for a whole hour.

LA40 + LB40 · Question 6: Normal, processing time (mean 50 ms, variance 15)

Known: X = waktu pemrosesan (ms), X ~ N(50, 15). **The variance is 15** (the question writes "varians 15 milidetik"; the unit should be ms^2), so **σ = √15 = 3.8730 ms**.
Method for every part: $z = \dfrac{x - 50}{3.873}$, round z to 2 decimals, read Φ(z) from Tab T.

a. P(X < 45)

WORKING $z = \dfrac{45 - 50}{3.873} = -1.29$ → Φ(-1.29)

ANSWER **0.0985**

CHECK Normal CD: Lower -1×10^{99}, Upper 45, σ = 3.873, μ = 50 → 0.09835

b. P(48 < X < 56)

WORKING $z_1 = \dfrac{48 - 50}{3.873} = -0.52$, $z_2 = \dfrac{56 - 50}{3.873} = 1.55$
Φ(1.55) - Φ(-0.52) = 0.9394 - 0.3015

ANSWER **0.6379** (Normal CD gives 0.6365; the difference comes from rounding z)

c. P(X > 60)

WORKING $z = \dfrac{60 - 50}{3.873} = 2.58$ → 1 - Φ(2.58) = 1 - 0.9951

ANSWER **0.0049**

TRAP With σ = 15 you would get z = 0.67 and 0.2514. A tail probability that large for a value only 10 ms above the mean should make you check σ.

d. Maximum time so that 97% of tasks finish within it

METHOD "Dalam waktu tersebut atau lebih cepat" = at most x, so the **left area is 0.97**. Find z, then $x = \mu + z\sigma$.

WORKING In the table body, 0.9699 (z = 1.88) is closest to 0.97 → z = 1.88
$x = 50 + 1.88(3.873) = 50 + 7.281$

ANSWER **x ≈ 57.28 ms**

CHECK Inverse Normal: Area 0.97, σ = 3.873, μ = 50 → 57.284

Sketch for every normal part: draw the bell curve, mark μ = 50, shade the asked region. It shows the marker which tail you used and catches "1 -" errors.

M · Practice Midterm: Probabilitas & Statistika

Format: same order and style as Quiz 1. Questions 1–6 are the core (about 90 minutes). Questions 7–8 are bonus questions on sessions 9–12, which Quiz 1 did not test. Open binder, calculator allowed. **Do not look at the answers (p.43) until you finish.**

1. Variabel dan skala pengukuran [12]

Seorang mahasiswa Computer Engineering meneliti kinerja jaringan IoT di sebuah gedung pintar. Variabel yang dicatat: **merek router, bandwidth (Mbps), latensi (ms), jumlah perangkat yang terhubung, kualitas sinyal (lemah/sedang/kuat), suhu ruang server (°C), status koneksi (terhubung/terputus), tingkat kepuasan pengguna (skala 1–5), packet loss (%), tahun pembuatan perangkat, nomor port, sisa daya baterai sensor (%)**. Tentukan jenis variabel dan skala pengukuran masing-masing variabel, beserta alasannya.

For each of the 12 variables, give the type, the measurement scale, and a one-line reason.

2. Statistika deskriptif [20]

Latensi jaringan (ms) diukur pada 15 waktu berbeda:

12, 15, 11, 48, 14, 13, 16, 12, 18, 15, 14, 52, 13, 17, 15

a. Hitung mean, median, dan modus.
b. Tentukan five number summary dan IQR.
c. Hitung varians dan standar deviasi sampel.
d. Hitung coefficient of variation.
e. Buatlah diagram stem and leaf.
f. Gambarkan boxplot dan identifikasi outlier.
g. Router kedua memiliki rata-rata latensi 30 ms dengan standar deviasi 3 ms. Router mana yang lebih stabil? Jelaskan dengan CV.
h. Hapus outlier, lalu hitung kembali mean dan CV. Apa yang dapat disimpulkan?

(a) Mean, median, mode. (b) Five-number summary, IQR. (c) Sample variance and SD. (d) CV. (e) Stem-and-leaf. (f) Boxplot and outliers. (g) A second router: mean 30 ms, SD 3 ms. Which is more stable? Use CV. (h) Remove the outliers and recompute the mean and CV.

3. Tabel kontingensi [16]

Kualitas koneksi 160 router dari tiga merek dicatat sebagai berikut:

Merek	Stabil	Kadang putus	Sering putus
A	45	10	5
B	30	20	10
C	15	10	15

a. Peluang router bermerek B dan sering putus.
b. Peluang router bermerek A atau sering putus.
c. Jika diketahui router sering putus, berapa peluang router tersebut bermerek C?
d. Jika diketahui router bermerek B, berapa peluang router tersebut kadang putus?
e. Jika diketahui router stabil, berapa peluang router tersebut bermerek A?
f. Apakah merek router dan kualitas koneksi saling independen? Tunjukkan perhitungannya.
g. Sebuah router yang sering putus dipilih secara acak. Merek apa yang paling mungkin?

(a) P(B and often drops). (b) P(A or often drops). (c) P(C | often drops). (d) P(sometimes drops | B). (e) P(A | stable). (f) Independent? Show the calculation. (g) A router that often drops: which brand is most likely?

4. Distribusi Binomial [14]

Sebuah pabrik chip mengetahui bahwa 10% chip yang diproduksi cacat. Seorang teknisi memilih 15 chip secara acak. Hitunglah peluang:

a. tepat 2 chip cacat
b. paling banyak 1 chip cacat
c. setidaknya 3 chip cacat
d. banyaknya chip cacat berada dari 1 hingga 3
e. sedikitnya satu chip cacat
f. Tentukan rata-rata dan standar deviasi banyaknya chip cacat.

10% of chips are defective; 15 chosen at random. (a) exactly 2 (b) at most 1 (c) at least 3 (d) from 1 to 3 (e) at least one (f) mean and SD.

5. Distribusi Poisson [14]

Sebuah web server menerima rata-rata 6 permintaan (request) per menit. Hitunglah peluang:

a. tepat 4 request dalam 1 menit
b. paling banyak 2 request dalam 1 menit
c. setidaknya 2 request dalam 1 menit
d. tidak ada request selama 20 detik
e. setidaknya 1 request selama 30 detik
f. Tentukan rata-rata dan standar deviasi banyaknya request per menit.

On average 6 requests per minute. (a) exactly 4 in a minute (b) at most 2 (c) at least 2 (d) none in 20 seconds (e) at least 1 in 30 seconds (f) mean and SD per minute.

6. Distribusi Normal [14]

Daya tahan baterai sebuah perangkat IoT berdistribusi normal dengan rata-rata 120 menit dan varians 64 menit^2. Hitunglah:

a. peluang baterai bertahan kurang dari 110 menit
b. peluang baterai bertahan antara 114 dan 130 menit
c. peluang baterai bertahan lebih dari 136 menit
d. daya tahan minimum agar sebuah baterai termasuk 5% baterai terbaik
e. batas waktu sehingga 90% baterai sudah habis pada waktu tersebut atau lebih cepat

Battery life ~ Normal, mean 120 min, variance 64. (a) P(X < 110) (b) P(114 < X < 130) (c) P(X > 136) (d) the minimum life to be in the best 5% (e) the time by which 90% of batteries have run out.

7. (Bonus) Variabel random kontinu [+5]

Diketahui $f(x) = \begin{cases} k(4-x) & 0 \le x \le 4 \\ 0 & x \text{ lainnya} \end{cases}$, dengan X = waktu respons sensor (detik).

a. Tentukan k agar f(x) merupakan pdf, lalu buktikan.
b. Hitung P(X ≤ 1), P(1 < X < 3), dan P(X > 2).
c. Tentukan fungsi kumulatif F(x).
d. Hitung μ dan σ^2.
e. Tentukan median.

(a) Find k and prove f is a pdf. (b) P(X ≤ 1), P(1 < X < 3), P(X > 2). (c) The cdf. (d) μ and σ^2. (e) The median.

8. (Bonus) Distribusi eksponensial [+5]

Waktu antar kegagalan sebuah hard disk berdistribusi eksponensial dengan rata-rata 200 jam. Hitunglah:

a. peluang gagal sebelum 100 jam
b. peluang bertahan lebih dari 300 jam
c. peluang gagal antara 150 dan 250 jam
d. peluang bertahan lebih dari 500 jam jika diketahui sudah bertahan 200 jam
e. median waktu kegagalan
f. waktu sehingga 90% hard disk sudah gagal

Exponential, mean 200 h. (a) $P(X < 100)$ (b) $P(X > 300)$ (c) $P(150 < X < 250)$ (d) $P(X > 500 \mid X > 200)$ (e) the median (f) the time by which 90% have failed.

— Selesai — Core 90 points + bonus 10.

M · Practice Midterm: worked answers

1. Variable type and measurement scale

Variabel	Jenis	Skala	Alasan
Merek router	Kualitatif	Nominal	labels with no order
Bandwidth (Mbps)	Kuantitatif kontinu	Rasio	0 Mbps = no bandwidth
Latensi (ms)	Kuantitatif kontinu	Rasio	measured time, true zero
Jumlah perangkat terhubung	Kuantitatif diskrit	Rasio	a count
Kualitas sinyal (lemah/sedang/kuat)	Kualitatif	Ordinal	ordered words, unequal gaps
Suhu ruang server (°C)	Kuantitatif kontinu	Interval	0°C is not "no heat"
Status koneksi	Kualitatif	Nominal	two labels, no order
Tingkat kepuasan (1–5)	Kualitatif	Ordinal	a ranking; the gap 1→2 need not equal 4→5
Packet loss (%)	Kuantitatif kontinu	Rasio	0% = no loss
Tahun pembuatan	Kuantitatif diskrit	Interval	year 0 is an arbitrary origin; 2024 is not "twice" 1012
Nomor port (80, 443 …)	Kualitatif	Nominal	numbers used as labels; averaging them is meaningless
Sisa daya baterai (%)	Kuantitatif kontinu	Rasio	0% = empty

The three that most often lose marks: **suhu °C** (interval), **tahun** (interval), and **nomor port** (nominal, even though it is a number).

2. Descriptive statistics (latency, ms)

Step 0: sort (bold = Q1, median, Q3)

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value	11	12	12	**13**	13	14	14	**15**	15	15	16	**17**	18	48	52

$n = 15 \cdot \Sigma x = 285 \cdot \Sigma x^2 = 7691$

Positions for n = 15: Q1 = 4th, median = 8th, Q3 = 12th.

a. Mean, median, mode

WORKING $\bar{x} = 285/15 \cdot$ median $= x_{(8)} \cdot$ mode = the value appearing most often (15 appears 3 times)

ANSWER **$\bar{x}$ = 19 ms, median = 15 ms, mode = 15 ms**

b. Five-number summary, IQR

ANSWER **Min 11, Q1 13, Median 15, Q3 17, Max 52 · IQR = 17 - 13 = 4**

c. Variance and SD

WORKING $s^2 = \dfrac{7691 - 285^2/15}{14} = \dfrac{7691 - 5415}{14} = \dfrac{2276}{14}$

ANSWER **$s^2 = 162.5714$ ms², $s = 12.7504$ ms**

CHECK STAT → sx = 12.7504

d. CV

WORKING CV = 12.7504 / 19 × 100%

ANSWER **CV = 67.11%**, very high variability

e. Stem-and-leaf

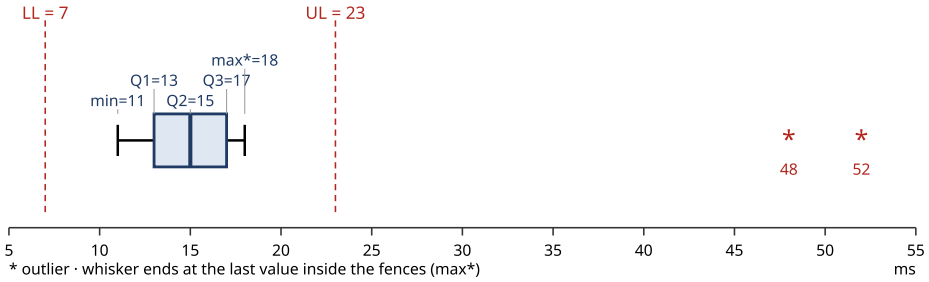
ANSWER

```
1 | 1 2 2 3 3 4 4 5 5 5 6 7 8
2 |
3 |
4 | 8
5 | 2
Key: 1 | 1 = 11 ms
```

f. Boxplot, outliers

WORKING LL = 13 - 1.5(4) = 7
UL = 17 + 1.5(4) = 23
48, 52 > 23. Whiskers: 11 to 18.

ANSWER **Outliers: 48 and 52 ms**



g. Which router is more stable?

WORKING Router 1: CV = 67.11%. Router 2: CV = 3/30 × 100% = 10%.

ANSWER **Router 2 is more stable**: its CV is smaller. Comparing SDs alone is not enough when the means differ.

h. Without the outliers (n = 13)

WORKING Σx = 285 - 48 - 52 = 185 → $\bar{x}$ = 14.2308 · STAT on the 13 values → sx = 2.0475 · CV = 2.0475/14.2308 × 100%

ANSWER **$\bar{x}$ = 14.23 ms, CV = 14.39%**. Two spikes caused most of the variability. The median (15) hardly changes, which shows it is robust.

3. Contingency table (router brand × connection quality)

	Stabil	Kadang	Sering	Total
A	45	10	5	**60**
B	30	20	10	**60**
C	15	10	15	**40**
Total	**90**	**40**	**30**	**160**

a. P(B ∩ Sering)

ANSWER 10/160 = **0.0625**

b. P(A ∪ Sering)

WORKING (60 + 30 - 5)/160 = 85/160

ANSWER **0.5313**

c. P(C | Sering)

WORKING divide by the Sering total: 15/30

ANSWER **0.5**

d. P(Kadang | B)

WORKING divide by the B total: 20/60

ANSWER **0.3333**

e. P(A | Stabil)

WORKING 45/90

ANSWER **0.5**

f. Independent?

WORKING P(Stabil | A) = 45/60 = 0.75, but P(Stabil) = 90/160 = 0.5625.
Equivalently P(A ∩ Stabil) = 45/160 = 0.2813 ≠ P(A)P(Stabil) = 0.375 × 0.5625 = 0.2109.

ANSWER **Not independent.** Brand A routers are stable more often than routers overall.

g. Most likely brand of a router that often drops

WORKING P(A | Sering) = 5/30 = 0.1667 · P(B | Sering) = 10/30 = 0.3333 · P(C | Sering) = 15/30 = 0.5

ANSWER **Brand C** (0.5). This is Bayes' rule read straight from the table.

4. Binomial: X ~ B(15, 0.1)

X = banyaknya chip cacat. P(0) = 0.9^{15} = 0.2059, P(1) = 0.3432, P(2) = 0.2669, P(3) = 0.1285, P(4) = 0.0428

a. P(X = 2)

WORKING $\binom{15}{2}(0.1)^2(0.9)^{13} = 105(0.01)(0.2542)$

ANSWER **0.2669**

b. P(X ≤ 1)

WORKING 0.2059 + 0.3432

ANSWER **0.5490**

c. P(X ≥ 3)

WORKING 1 - CD(2) = 1 - 0.8159 (CD(2) = 0.2059 + 0.3432 + 0.2669, unrounded 0.81594)

ANSWER **0.1841**

d. P(1 ≤ X ≤ 3)

WORKING 0.3432 + 0.2669 + 0.1285 = CD(3) - CD(0)

ANSWER **0.7386**

e. P(X ≥ 1)

WORKING $1 - 0.9^{15}$

ANSWER **0.7941**

f. Mean and SD

WORKING $\mu = 15(0.1) \cdot \sigma = \sqrt{(15 \cdot 0.1 \cdot 0.9)} = \sqrt{1.35}$

ANSWER **$\mu = 1.5, \sigma = 1.1619$**

5. Poisson: 6 requests per minute

X = request per menit, X ~ Poisson(6). e^{-6} = 0.0024788. P(0) = 0.0025, P(1) = 0.0149, P(2) = 0.0446, P(3) = 0.0892, P(4) = 0.1339

a. P(X = 4)

WORKING $\frac{e^{-6}6^4}{4!} = \frac{0.0024788 \times 1296}{24}$

ANSWER **0.1339**

b. P(X ≤ 2)

WORKING $e^{-6}(1 + 6 + 18) = 25e^{-6}$

ANSWER **0.0620**

c. P(X ≥ 2)

WORKING 1 - P(0) - P(1) = 1 - $7e^{-6}$

ANSWER **0.9826**

d. No request in 20 seconds

METHOD Rescale: 20 s = 1/3 minute → λ = 6 × 1/3 = 2

ANSWER e^{-2} = **0.1353**

e. At least 1 in 30 seconds

METHOD λ = 6 × 0.5 = 3; P(Y ≥ 1) = 1 - e^{-3}

ANSWER **0.9502**

f. Mean and SD per minute

WORKING For a Poisson, mean = variance = λ

ANSWER **μ = 6, σ = √6 = 2.4495**

6. Normal: X ~ N(120, 64), σ = √64 = 8

a. P(X < 110)

WORKING z = (110 - 120)/8 = -1.25 → Φ(-1.25)

ANSWER **0.1056**

b. P(114 < X < 130)

WORKING z = -0.75 and 1.25 → 0.8944 - 0.2266

ANSWER **0.6678**

c. P(X > 136)

WORKING z = 16/8 = 2.00 → 1 - 0.9772

ANSWER **0.0228**

d. Minimum life to be in the best 5%

METHOD Best 5% = right tail 0.05 → left area 0.95 → z = 1.645

WORKING x = 120 + 1.645(8)

ANSWER **133.16 minutes** (Inverse Normal 0.95 → 133.159)

e. Time by which 90% have run out

METHOD "90% sudah habis pada waktu tersebut atau lebih cepat" = left area 0.90 → z = 1.28

WORKING x = 120 + 1.28(8)

ANSWER **130.24 minutes** (Inverse Normal → 130.25)

7. (Bonus) Continuous pdf: f(x) = k(4 - x) on [0, 4]

a. Find k and prove f is a pdf

WORKING $\int_0^4 k(4-x)\,dx = k\left[4x - \frac{x^2}{2}\right]_0^4 = k(16-8) = 8k = 1 \Rightarrow k = \frac{1}{8}$.

For 0 ≤ x ≤ 4, 4 - x ≥ 0, so f(x) ≥ 0. Elsewhere f = 0. Both conditions hold ∴ f is a pdf.

ANSWER **k = 1/8 = 0.125**

b. Probabilities

WORKING Use $F(x) = \frac{1}{8}\left(4x - \frac{x^2}{2}\right)$: F(1) = 3.5/8 = 0.4375 · F(3) = 7.5/8 = 0.9375 · F(2) = 6/8 = 0.75

ANSWER **P(X ≤ 1) = 0.4375 · P(1 < X < 3) = 0.5 · P(X > 2) = 0.25**

c. cdf

ANSWER

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{8x - x^2}{16} & 0 \le x \le 4 \\ 1 & x > 4 \end{cases}$$

· Check: F(0) = 0 and F(4) = 1 ✓

d. μ and σ²

WORKING $\mu = \frac{1}{8}\int_0^4 (4x - x^2)dx = \frac{1}{8}\left(32 - \frac{64}{3}\right) = \frac{4}{3}$ · $\mathrm{E}(X^2) = \frac{1}{8}\int_0^4 (4x^2 - x^3)dx = \frac{1}{8}\left(\frac{256}{3} - 64\right) = \frac{8}{3}$

σ² = 8/3 - (4/3)² = 8/9

ANSWER **μ = 1.3333 s, σ² = 0.8889 s²**

e. Median

WORKING F(m) = 0.5 → 8m - m² = 8 → m² - 8m + 8 = 0 → m = 4 ± √8. Only 4 - √8 lies in [0, 4].

ANSWER **m = 1.1716 s** (median < mean: the density decreases, so the distribution is skewed right)

8. (Bonus) Exponential: mean 200 h → λ = 1/200

a–c

WORKING P(X < 100) = 1 - e^(-100/200) · P(X > 300) = e^(-1.5) · P(150 < X < 250) = e^(-0.75) - e^(-1.25)

ANSWER **0.3935 · 0.2231 · 0.1859**

d. P(X > 500 | X > 200)

METHOD Memoryless: only the extra 300 hours count.

ANSWER e^(-300/200) = **0.2231**, the same as (b)

e–f. Median and 90th percentile

WORKING Solve 1 - e^(-x/200) = p → x = -200 ln(1 - p). p = 0.5 → 200 ln 2. p = 0.9 → -200 ln 0.1.

ANSWER **median = 138.63 h · 90% by 460.52 h**

Mark yourself

Lost marks on Q1	Re-read the scale decision steps (p.8) and the traps on °C, years and port numbers.
Lost marks on Q2	Did you sort first? Did you use sx? Did you stop the whiskers at the last value inside the fences?
Lost marks on Q4–Q6	Check Tab X: 1 - CD(k - 1) for "at least", rescaling λ, σ = √variance, left area for "or faster".

A · Practice Set A (the v1 quiz)

Rules: open binder, calculator allowed, **answers start on p.52**. Suggested time is 100 minutes for all 16 questions, or about 50 minutes for any 8. Topics are mixed on purpose, so part of the task is deciding which method to use (Tab I and Tab D). Write X = ..., the distribution, the formula, the substitution and the answer to 4 decimals. Each question is in Indonesian with an English line underneath.

1. Data kelulusan 200 mahasiswa mata kuliah Probstat berdasarkan gender: [8]

	Lulus (L)	Tidak Lulus (TL)
Perempuan (F)	72	18
Laki-laki (M)	78	32

Jika seorang mahasiswa dipilih secara acak, hitunglah: (a) P(F ∩ L) (b) P(L) (c) P(L | F) (d) P(F | TL) (e) P(F ∪ L). (f) Apakah kejadian F dan L saling bebas? Jelaskan.

200 students by gender and pass/fail. Pick one at random. Find (a)–(e). (f) Are F and L independent? Explain.

2. Berat bersih kemasan kopi berdistribusi normal dengan rata-rata 500 gram dan varians 25 gram². [7] Tentukan peluang sebuah kemasan memiliki berat: (a) kurang dari 490 gram (b) antara 495 dan 510 gram (c) lebih dari 512 gram. (d) Tentukan berat minimum agar sebuah kemasan termasuk 10% kemasan terberat.

Net weight ~ Normal, mean 500 g, variance 25. Find (a) P(X < 490), (b) P(495 < X < 510), (c) P(X > 512), (d) the weight that marks the heaviest 10%.

3. Distribusi peluang variabel random X diberikan sebagai berikut: [7]

x	0	1	2	3
P(X = x)	0,1	k	2k	0,3

(a) Tentukan nilai k. (b) Hitung P(X ≥ 2) dan F(1,5). (c) Hitung E(X) dan Var(X). (d) Jika Y = 3X - 2, hitung E(Y) dan Var(Y).

(a) Find k. (b) P(X ≥ 2) and F(1.5). (c) E(X), Var(X). (d) For Y = 3X - 2, find E(Y) and Var(Y).

4. Waktu tunggu (menit) 12 pelanggan di sebuah kafe: 4, 7, 5, 9, 6, 12, 5, 8, 30, 6, 7, 5. Hitunglah: (a) rata-rata, median, dan modus (b) varians dan simpangan baku sampel (c) Q1, Q3, dan IQR (d) Apakah ada pencilan? Gunakan batas Q1 - 1,5·IQR dan Q3 + 1,5·IQR. (e) Bagaimana bentuk distribusinya, dan ukuran pemusatan mana yang lebih tepat? [8]

Waiting times of 12 customers. (a) Mean, median, mode. (b) Sample variance and SD. (c) Q1, Q3, IQR. (d) Any outliers (1.5·IQR fences)? (e) Shape, and which measure of centre is better?

5. Rata-rata terdapat 4 panggilan masuk per jam ke layanan helpdesk kampus. Hitunglah peluang: (a) tepat 2 panggilan dalam 1 jam (b) paling banyak 3 panggilan dalam 1 jam (c) paling sedikit 1 panggilan dalam 30 menit (d) lebih dari 5 panggilan dalam 1 jam. [6]

On average 4 calls per hour. (a) Exactly 2 in an hour. (b) At most 3 in an hour. (c) At least 1 in 30 minutes. (d) More than 5 in an hour.

6. Diketahui fungsi $f(x) = \begin{cases} c(x+1) & 1 \le x \le 3 \\ 0 & x \text{ lainnya} \end{cases}$. (a) Tentukan c agar f(x) merupakan pdf. (b) Hitung P(X ≤ 2) dan P(1,5 < X < 2,5). (c) Tentukan fungsi kumulatif F(x). (d) Hitung μ dan σ². (e) Tentukan median X. [8]

(a) Find c so f is a pdf. (b) P(X ≤ 2), P(1.5 < X < 2.5). (c) The cdf F(x). (d) μ and σ². (e) The median.

7. Sebuah pabrik memiliki tiga mesin A, B, dan C yang masing-masing menghasilkan 50%, 30%, dan 20% dari total produksi. Persentase produk cacat dari mesin A, B, dan C berturut-turut adalah 2%, 3%, dan 5%. (a) Berapa peluang sebuah produk yang diambil secara acak cacat? (b) Jika produk yang diambil ternyata cacat, berapa peluang produk tersebut berasal dari mesin C? (c) Mesin mana yang paling mungkin menghasilkan produk cacat tersebut? [7]

Machines A, B, C make 50%, 30%, 20% of output, with defect rates 2%, 3%, 5%. (a) P(defective). (b) P(C | defective). (c) Which machine most likely produced a given defective item?

8. Peluang sebuah komponen elektronik cacat adalah 0,1. Diambil 12 komponen secara acak (saling bebas). Hitunglah peluang: (a) tepat 2 komponen cacat (b) paling banyak 1 komponen cacat (c) paling sedikit 3 komponen cacat (d) sedikitnya satu komponen cacat. (e) Hitung rata-rata dan simpangan baku banyaknya komponen cacat. [6]

P(defective) = 0.1, 12 independent components. (a) Exactly 2. (b) At most 1. (c) At least 3. (d) At least one. (e) Mean and SD.

9. Umur sebuah lampu LED berdistribusi eksponensial dengan rata-rata 1000 jam. Hitunglah: (a) peluang lampu mati sebelum 500 jam (b) peluang lampu bertahan lebih dari 1500 jam (c) peluang lampu bertahan lebih dari 1500 jam jika diketahui lampu sudah menyala 1000 jam (d) median umur lampu. [6]

LED lifetime ~ Exponential, mean 1000 h. (a) P(fails before 500 h). (b) P(lasts more than 1500 h). (c) P(lasts more than 1500 h | it has already lasted 1000 h). (d) Median lifetime.

10. (a) Dari 6 pria dan 4 wanita akan dibentuk panitia beranggotakan 4 orang. Berapa banyak cara membentuk panitia? Berapa peluang panitia terdiri dari tepat 2 wanita? Berapa peluang paling sedikit 1 wanita? (b) Dari 5 kandidat akan dipilih ketua, sekretaris, dan bendahara. Ada berapa cara? [6]

(a) A committee of 4 from 6 men and 4 women: number of ways, P(exactly 2 women), P(at least 1 woman). (b) Choose a chair, secretary and treasurer from 5 candidates: number of ways.

11. Sebuah lab memiliki 15 laptop, 4 di antaranya rusak. Seorang asisten mengambil 5 laptop secara acak **tanpa pengembalian**. Misalkan X = banyaknya laptop rusak yang terambil. (a) Hitung P(X = 1). (b) Hitung peluang paling sedikit satu laptop rusak terambil. (c) Hitung E(X) dan Var(X). [6]

15 laptops, 4 broken. Pick 5 without replacement; X = number broken. (a) P(X = 1). (b) P(at least one broken). (c) E(X), Var(X).

12. Buktikan bahwa fungsi $f(x) = \begin{cases} 0{,}2\, e^{-0{,}2(x-1)} & x \geq 1 \\ 0 & x \text{ lainnya} \end{cases}$ merupakan sebuah pdf. Kemudian hitung P(X > 6) dan P(2 < X < 4). [6]

Prove that f is a pdf. Then find P(X > 6) and P(2 < X < 4).

13. Nilai UTS suatu kelas memiliki rata-rata 70 dan simpangan baku 8. (a) Tanpa mengetahui bentuk distribusinya, paling sedikit berapa proporsi mahasiswa yang nilainya antara 54 dan 86? (b) Jika distribusinya berbentuk lonceng (mound-shaped), kira-kira berapa proporsi mahasiswa yang nilainya di atas 86? (c) ... dan antara 62 dan 86? [5]

Midterm scores: mean 70, SD 8. (a) With no shape assumption, at least what proportion is between 54 and 86? (b) If mound-shaped, roughly what proportion is above 86? (c) Between 62 and 86?

14. Diketahui 30% pengguna aplikasi membuka fitur premium. Diambil sampel 100 pengguna. Dengan pendekatan distribusi normal, hitunglah peluang (a) paling banyak 25 pengguna membuka fitur premium (b) tepat 30 pengguna. (c) Bandingkan dengan nilai eksak Binomial (gunakan kalkulator). [6]

30% use the premium feature, n = 100. Using the normal approximation: (a) P(at most 25). (b) P(exactly 30). (c) Compare with the exact Binomial.

15. Bus kampus datang setiap 20 menit. Waktu tunggu seorang mahasiswa yang datang secara acak berdistribusi seragam pada [0, 20] menit. Hitung (a) peluang menunggu lebih dari 15 menit (b) peluang menunggu antara 5 dan 12 menit (c) rata-rata dan varians waktu tunggu. [4]

Wait ~ Uniform[0, 20] min. (a) P(W > 15). (b) P(5 < W < 12). (c) Mean and variance.

16. Sebuah gedung memiliki 3 alarm asap yang bekerja saling bebas. Masing-masing mendeteksi asap dengan peluang 0,85. (a) Berapa peluang paling sedikit satu alarm berbunyi saat terjadi kebakaran? (b) Berapa peluang tepat dua alarm berbunyi? (c) Apakah kejadian "alarm 1 berbunyi" dan "alarm 2 berbunyi" saling lepas? Jelaskan. [4]

3 independent smoke alarms, each detects with probability 0.85. (a) P(at least one sounds). (b) P(exactly two). (c) Are "alarm 1 sounds" and "alarm 2 sounds" mutually exclusive? Explain.

— Selesai (end) — Total 100 points. Answers on **p.52**.

A · Practice Set A: worked answers

The "→ p." links point to the binder page with the method. Answers are given to 4 decimals. A z-table answer may differ from Normal CD in the 3rd–4th decimal, and both are accepted.

1. Contingency table → p.12

Totals: F = 90, M = 110, L = 150, TL = 50, N = 200.

(a) P(F ∩ L) = 72/200 = **0.36** (b) P(L) = 150/200 = **0.75** (c) P(L | F) = 72/90 = **0.80** (d) P(F | TL) = 18/50 = **0.36**

(e) P(F ∪ L) = (90 + 150 - 72)/200 = 168/200 = **0.84**

(f) P(L | F) = 0.80 ≠ P(L) = 0.75. Equivalently, P(F)P(L) = 0.45 × 0.75 = 0.3375 ≠ 0.36. So F and L are **not independent**: female students pass at a higher rate.

2. Normal → p.20, p.21

X ~ N(500, 25), so **σ = 5** (not 25!).

(a) z = (490 - 500)/5 = -2.00 → Φ(-2.00) = **0.0228**

(b) z = -1.00 and 2.00 → 0.9772 - 0.1587 = **0.8185**

(c) z = 12/5 = 2.40 → 1 - 0.9918 = **0.0082**

(d) Heaviest 10% means a left area of 0.90, so z = 1.28 and x = 500 + 1.28(5) = **506.4 g** (Inverse Normal: 506.41)

3. Discrete table → p.14, p.14

(a) 0.1 + k + 2k + 0.3 = 1 → **k = 0.2**, so the table is 0.1, 0.2, 0.4, 0.3.

(b) P(X ≥ 2) = 0.4 + 0.3 = **0.7**; F(1.5) = P(X ≤ 1.5) = P(0) + P(1) = **0.3**

(c) E(X) = 0 + 0.2 + 0.8 + 0.9 = **1.9**; $E(X^2)$ = 0 + 0.2 + 1.6 + 2.7 = 4.5; Var(X) = 4.5 - 1.9^2 = **0.89**

(d) E(Y) = 3(1.9) - 2 = **3.7**; Var(Y) = 3^2(0.89) = **8.01** (the -2 does not change the variance)

4. Descriptive → p.8, p.9

Sorted: 4 5 5 5 6 6 7 7 8 9 12 30 (n = 12), Σx = 104, Σx^2 = 1450.

(a) $\bar{x}$ = 104/12 = **8.6667**; median position 6.5 → (6 + 7)/2 = **6.5**; mode = **5**

(b) $s^2 = \frac{1450 - 104^2/12}{11} = \frac{548.667}{11} =$ **49.8788**, s = **7.0625** (STAT: sx)

(c) Q1 position 0.25(13) = 3.25 → 5 + 0.25(5 - 5) = **5**; Q3 position 9.75 → 8 + 0.75(9 - 8) = **8.75**; IQR = **3.75**

(d) LL = 5 - 5.625 = -0.625, UL = 8.75 + 5.625 = 14.375, so **30 is an outlier**

(e) Mean > median, so the distribution is **skewed right** because of the outlier. The **median** is the better measure of centre: without the 30, $\bar{x}$ = 6.73 but the median barely moves, to 6.

5. Poisson → p.15

X = number of calls per hour, X ~ Poisson(4).

(a) P(X = 2) = $e^{-4}4^2/2!$ = **0.1465**

(b) P(X ≤ 3) = e^{-4}(1 + 4 + 8 + 10.667) = 0.0183 + 0.0733 + 0.1465 + 0.1954 = **0.4335**

(c) 30 minutes, so **λ = 2**: P(Y ≥ 1) = 1 - e^{-2} = **0.8647**

(d) P(X > 5) = 1 - P(X ≤ 5) = 1 - 0.7851 = **0.2149** (Poisson CD x = 5, μ = 4)

6. General pdf → p.17, p.17, p.17

(a) $\int_1^3 c(x+1)dx = c\left[\frac{x^2}{2} + x\right]_1^3 = c(7.5 - 1.5) = 6c = 1 \Rightarrow$ **c = 1/6**. f ≥ 0 on [1, 3] ✓

(b) $\mathrm{P}(X \le 2) = \frac{1}{6}\left[\frac{x^2}{2} + x\right]_1^2 = \frac{1}{6}(4 - 1.5) =$ **0.4167**; P(1.5 < X < 2.5) = F(2.5) - F(1.5) = 0.6875 - 0.1875 = **0.5**

(c) $F(x) = \begin{cases} 0 & x < 1 \\ \frac{x^2 + 2x - 3}{12} & 1 \le x \le 3 \\ 1 & x > 3 \end{cases}$. Check: F(1) = 0 and F(3) = 12/12 = 1 ✓

(d) $\mu = \frac{1}{6}\int_1^3 (x^2 + x)dx = \frac{1}{6}\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_1^3 = \frac{1}{6}\left(13.5 - \frac{5}{6}\right) = \frac{19}{9} =$ **2.1111**

$\mathrm{E}(X^2) = \frac{1}{6}\left[\frac{x^4}{4} + \frac{x^3}{3}\right]_1^3 = \frac{1}{6}\left(29.25 - \frac{7}{12}\right) = \frac{43}{9} = 4.7778$; σ^2 = 43/9 - $(19/9)^2$ = 26/81 = **0.3210**

(e) F(m) = 0.5 → m^2 + 2m - 3 = 6 → m^2 + 2m - 9 = 0 → m = -1 + √10 = **2.1623** (take the root inside [1, 3])

7. Total probability + Bayes → p.13

(a) P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = **0.029**

(b) P(C | D) = 0.010/0.029 = **0.3448**

(c) P(A | D) = 0.3448, P(B | D) = 0.3103, P(C | D) = 0.3448. **A and C are tied** as the most likely source. A has the lowest defect rate but makes the most products. Check: the three add to 1.

8. Binomial → p.15

X = number of defective components, X ~ B(12, 0.1).

(a) $\binom{12}{2}(0.1)^2(0.9)^{10} = 66(0.01)(0.3487) =$ **0.2301**

(b) P(0) + P(1) = 0.2824 + 0.3766 = **0.6590**

(c) P(X ≥ 3) = 1 - P(X ≤ 2) = 1 - (0.6590 + 0.2301) = **0.1109**

(d) 1 - P(0) = 1 - 0.9^{12} = **0.7176**

(e) μ = 12(0.1) = **1.2** ; σ = √(12 · 0.1 · 0.9) = √1.08 = **1.0392**

9. Exponential → p.22, p.22

Mean 1000, so λ = 1/1000 = 0.001 per hour.

(a) P(X < 500) = 1 - e^(-0.5) = **0.3935** (b) P(X > 1500) = e^(-1.5) = **0.2231**

(c) Memoryless: P(X > 1500 | X > 1000) = P(X > 500) = e^(-0.5) = **0.6065**

(d) Median = ln 2/λ = **693.15 h**

10. Counting → p.11

(a) Ways = C(10, 4) = **210** . P(exactly 2 women) = C(4,2)·C(6,2)/210 = 6·15/210 = **0.4286** . P(at least 1 woman) = 1 - C(6,4)/210 = 1 - 15/210 = **0.9286**

(b) The roles are different, so order matters: P(5, 3) = 5·4·3 = **60**

11. Hypergeometric → p.16

N = 15, M = 4 broken, n = 5.

(a) $P(X=1) = \frac{\binom{4}{1}\binom{11}{4}}{\binom{15}{5}} = \frac{4\cdot 330}{3003} =$ **0.4396**

(b) 1 - P(X = 0) = 1 - C(11,5)/C(15,5) = 1 - 462/3003 = **0.8462**

(c) E(X) = 5(4/15) = **1.3333** ; Var(X) = (10/14)(5)(4/15)(11/15) = **0.6984**

12. Prove pdf → p.17

(1) For x ≥ 1, 0.2 > 0 and e^(·) > 0, so f(x) > 0. Elsewhere f = 0. So f ≥ 0 ✓

(2) $\int_{-\infty}^{\infty} f = \int_1^{\infty} 0.2e^{-0.2(x-1)}dx = \left[-e^{-0.2(x-1)}\right]_1^{\infty} = 0-(-1) = 1$ ✓ ∴ f is a pdf.

P(X > 6) = $\left[-e^{-0.2(x-1)}\right]_6^{\infty} = e^{-1} =$ **0.3679** ; P(2 < X < 4) = e^(-0.2) - e^(-0.6) = 0.8187 - 0.5488 = **0.2699**

This is an exponential distribution with λ = 0.2 shifted to start at 1, so E(X) = 1 + 5 = 6.

13. Chebyshev and empirical rule → p.9

(a) 54 and 86 are 70 ± 16 = 70 ± 2(8), so k = 2: at least 1 - 1/4 = **0.75**

(b) 86 = μ + 2σ, so ≈ (1 - 0.95)/2 = **0.025**

(c) 62 = μ - σ, 86 = μ + 2σ: ≈ 0.68/2 + 0.95/2 = **0.815** (the z-table gives 0.8185)

14. Normal approximation → p.21

X ~ B(100, 0.3). np = 30 ≥ 5 and nq = 70 ≥ 5, so the approximation is OK. μ = 30, σ = √21 = 4.5826.

(a) P(X ≤ 25) ≈ P(Y ≤ 25.5): z = (25.5 - 30)/4.5826 = -0.98 → **0.1635**

(b) P(X = 30) ≈ P(29.5 ≤ Y ≤ 30.5): z = ±0.11 → 0.5438 - 0.4562 = **0.0876**

(c) Exact: Binomial CD(25; 100, 0.3) = **0.1631**, Binomial PD(30) = **0.0868**. The approximation is very close.

15. Uniform → p.19

(a) (20 - 15)/20 = **0.25** (b) (12 - 5)/20 = **0.35** (c) μ = 10 min, $\sigma^2 = 20^2/12$ = **33.33**

16. Independence → p.11, p.13

(a) 1 - P(none) = 1 - 0.15^3 = **0.9966**

(b) B(3, 0.85): $3(0.85)^2(0.15) =$ **0.3251**
(c) **No.** They can happen together: P(both) = $0.85^2 = 0.7225 \neq 0$. The alarms are independent but not mutually exclusive.

Self-score

Score	What to do next
Missed any σ vs σ^2, ≤ vs <, or λ-rescaling	Re-read Tab X before the exam. These are exactly the class-board traps.
Lost time choosing a method	Use Tab I first next time, before reading the chapter pages.
Calculation slips	Keep unrounded values in STO memory and run the DIST check values (Tab C) tonight.